

THE LOSS DOES NOT SEE THE BASIS, BUT ADAM DOES

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ABSTRACT

Gradient descent on a factored model $W = UV^\top$ is implicitly biased toward low-rank solutions, while Adam, starting from the same small initialization, is not. We trace the difference to the gauge symmetry of the loss, its invariance under $(U, V) \mapsto (UQ, VQ)$. Gradient flow’s low-rank mechanism is available to an optimizer only if that optimizer is gauge-equivariant, a condition necessary for the transfer but not sufficient for low-rank recovery. Gradient descent, momentum, “shared-scalar” Adam, Muon, and Shampoo satisfy it. Adam, RMSProp, and the other coordinate-wise methods do not. A structure theorem characterizes the memoryless equivariant rules as exactly the Gram-determined left preconditioners, and a transfer theorem carries gradient flow’s pathwise properties to common-scalar flows. We then sort nine update rules on underdetermined matrix sensing by recovery error against the planted ground truth. A one-parameter family from coordinate-wise to shared-scalar preconditioning restores the bias monotonically, isolating anisotropy as the cause. A “spectral schedule” reconciles two opposing reports about Muon: equal-rate updates recover exactly low-rank targets but lose their edge as the spectral tail grows. In transformers, Adam separates two gauge-equivalent initializations at the first step, where the equivariant optimizers stay at float precision, and ends with the per-head invariants $W_Q^\top W_K$ 56% apart in relative Frobenius distance, a gap no per-head rotation can close. On two hyperspectral datasets at matched training loss, gradient descent cuts held-out error by 43–44% at the lowest sampling density, and at lower effective rank. Basis choice is therefore not a tuning detail but a decision about which interpolant the optimizer selects.

1 INTRODUCTION

The implicit regularization effect of gradient descent (GD) in factored models is by now well established: training an overparameterized factorization to convergence, i.e. to a global minimum of an underdetermined matrix problem starting from small initializations, does not yield an arbitrary interpolating solution. In well-studied regimes it yields approximately the minimum-nuclear-norm one, a convex proxy which strongly promotes low rank (Gunasekar et al., 2017; Arora et al., 2019; Li et al., 2021). By contrast, empirical results show that adaptive methods have different generalization properties—Adam-family optimizers, while achieving a similar train loss, can end up measurably worse on the test set (Wilson et al., 2017; Keskar & Socher, 2017). Recent work showed that per-coordinate adaptivity is basis-dependent (Xie et al., 2025), while a preconditioner shared across coordinates restores rotational equivariance (Ling et al., 2022). We share this concern about basis dependence. To the best of our knowledge, its implication for interpolant selection on a factored model has not been characterized: whether there is a criterion by which one can determine that an optimizer preserves or breaks the low-rank inductive bias, in such a way that a range of deployed methods can be sorted by that criterion and the sorting verified against ground truth.

This is particularly consequential in the context of individual attention heads, which carry the same symmetry: two transformers that compute the identical function at initialization start behaving

§§1–11 are self-contained. The appendices contain the proofs and the control battery.

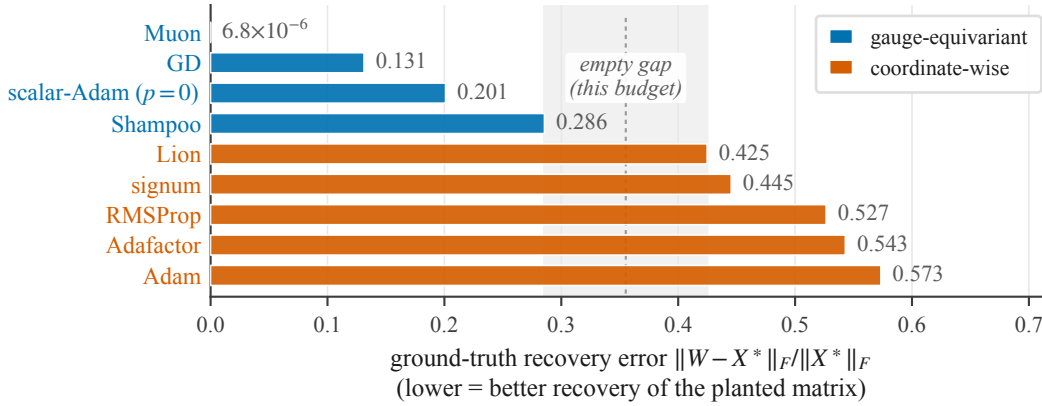


Figure 1: Equivariance structures the set of available optimizers under a fixed computational budget. Ground-truth recovery for nine optimizer rules (eight standard rules plus a shared-scalar Adam as a baseline) on an underdetermined matrix sensing task without weight decay. All are run until the training residuals vanish (interpolation), which rules out any performance difference due to residual training error. The four gauge-equivariant methods (blue) maintain the low-rank bias while the five coordinate-wise approaches (vermillion) do not: on this task and at this budget, recovery error is at most 0.286 for every equivariant method and above 0.42 for every coordinate-wise one, leaving the space between the two unoccupied. That qualifier matters, as one of the coordinate-wise methods eventually reaches similar performance beyond the 2×10^4 -step budget used here (Appendix D.10). Muon’s label reads 6.8×10^{-6} , the unrounded value of the 0.0000 entry in Table 2.

differently from the first Adam update, and ultimately exhibit head-level invariants 56% apart at matched validation accuracy (§6).

We answer this question by introducing a criterion according to which an optimizer can be said to preserve the implicit low-rank bias of factored models. The objective function $L(U, V) = f(UV^\top)$ remains unchanged under the transformation

$$(U, V) \mapsto (UQ, VQ), \quad Q \in O(k),$$

meaning that rotating the latent factor bases does not alter the resulting matrix $W = UV^\top$ or any model prediction. This property is referred to as the gauge symmetry. Optimization takes place in the space of (U, V) factorizations, and different algorithms handle this symmetry to varying extents. A central question is whether an optimizer is equivariant under orthogonal transformations of the gauge. The ones that are can exhibit the low-rank behavior typically seen in gradient flow on factored models.

Equivariance allows us to reproduce gradient-flow dynamics, but does not in general lead to low-rank solutions (see Remark 4.7). For equivariant optimizers, the key to low-rank convergence is the spectral schedule: whether the dynamics it induces on the singular values of the factored weights during training lead to attraction to a low-rank manifold within the specified training budget. For optimizers whose updates respect equivariance—gradient descent, momentum, shared-scalar-preconditioned Adam, Muon, Shampoo—we can say that their behavior is consistent across all gauge-equivalent factorizations, while for coordinate-wise algorithms—Adam, RMSProp, signSGD, Lion, Adafactor—each optimizer reads a preferred basis, so it can select a different interpolant depending on where in the gauge orbit it starts. In particular, in the optimization problems considered in this paper, the coordinate-wise optimizers failed to reach the low-rank attractor, settling instead on interpolants of higher rank and worse recovery (Figure 1, Table 2). Thus only equivariant algorithms were able to access the low-rank solutions found by gradient flow. Note, however, that this does not necessarily mean that a coordinate-wise optimizer cannot converge to a low-rank solution by some other route, just that it has less direct access to it. One example of such a case is given in Appendix D.10.

Contributions. We develop this criterion in five steps.

1. **Classification and first-step defect analysis (§4).** We characterize those standard optimization updates that are compatible with the gauge action. Gradient descent (GD), Polyak momentum, Adam with a shared scalar second moment, Muon’s matrix sign update (and all finite Newton–Schulz iterates of it), and Shampoo’s Kronecker preconditioner with damping

are all equivariant. By contrast, any optimizer that applies a fixed entrywise nonlinearity at zero optimizer state (Adam, RMSProp, signSGD, Lion) is not. Similar distinctions have been noted previously for unfactored parameterizations, where the rotation acts on the whole parameter space (§2, Table 1), but not for the internal gauge of a factored model. For Adam with zero-initialized state, we obtain an exact expression for the first-step defect. We then present a transfer theorem: optimization dynamics with a shared scalar preconditioner correspond to gradient flow under a rescaled time parameterization, hence previous results for gradient flow (Gunasekar et al., 2017; Arora et al., 2019; Li et al., 2021) carry over directly whenever their own conditions are met by the same objective. A structure theorem turns the classification into a full characterization: a memoryless update is equivariant exactly when it takes the form $\Delta = H(GG^\top)G$ for some preconditioner H , whose choice defines the optimizer (Theorem 4.5, stated for full-column-rank G with $k \leq n$). The defining property is equivariance, not balancedness—which both Muon and Shampoo violate while still preserving the bias.

2. **Optimizer comparison (§5).** In matrix sensing without weight decay, all optimizers are driven to interpolation, erasing residual training error as a source of variation. Under this controlled condition, the theory cleanly separates nine update rules (eight deployed optimizers plus a control, scalar-Adam) into two classes by their recovery error at this computational budget (equivariant 0.00–0.29; coordinate-wise 0.42–0.57; Figure 1). This distinction is robust to the various controls explored in Appendix D: symmetrized schedules (cosine decay for all methods), full learning-rate scans, and variable initialization amplitude. The predicted ranking of GD, scalar-Adam, and Muon at lower error than the coordinate-wise group is observed up to $n=256$ (10 seeds, Appendix D.9), with two flagged exceptions to the strict two-class split (untuned Shampoo and long-anneal signum; §11).
3. **A dial that isolates the mechanism (§7).** We construct a parameterized family of update rules which smoothly interpolates between preconditioner geometries as p ranges over $[0, 1]$. The interpolation connects Adam’s standard coordinate-wise denominator ($p=1$) to a single coordinate-independent scalar ($p=0$), which is gauge-equivariant. Along this interpolation, recovery and effective rank improve monotonically. The only thing that changes with p is the denominator’s coordinate-wise anisotropy—the part of the update that reads the gauge—plus the overall scaling that comes with it; adaptivity and momentum are kept constant throughout. The same trends hold when the learning rate is re-optimized at each p (§7).
4. **The same gauge lives in attention (§6).** Because each head’s logits depend on W_Q and W_K only through $W_Q^\top W_K$, each head has the same internal gauge symmetry as the factored model. We train two gauge-equivalent copies of a small transformer—the same function at initialization, written in different bases—to compare the behavior of Adam and heavy-ball SGD. Adam splits them at the first step, and the two copies end with head invariants $W_Q^\top W_K$ that differ by 56%. Heavy-ball SGD and scalar-Adam, by contrast, are indistinguishable to floating-point precision at that step and stay orders of magnitude below Adam’s split afterwards, and an $A=I$ twin confirms that the pipeline itself is exactly deterministic. A sanity check demonstrates that it is not some imprecision in the calculation that produces Adam’s divergence: perturbing the weights by 10^{-7} in the same basis leaves a step-1 drift four orders of magnitude below the gauge split. Only for Muon does the apparent gauge divergence come from numerical instability in the matrix sign calculation, and in particular from the presence of small singular values. This is not specific to the small transformer or to deterministic training: the same effect can be seen when training a six-layer character-level language model with $d_{\text{model}}=256$ under mini-batch noise, where the Adam twins again diverge at step one while the equivariant ones stay at machine precision.
5. **Inside the equivariant class, and on real data (§8–§10).** The optimizer sweep here has essentially two knobs: equivariance, which sets class membership and is what the dial of §7 moves along, and the aggressiveness of the spectral schedule, which sets behavior inside the class. We characterize the latter by a phase diagram in the tail energy, the fraction of the target’s energy carried outside its low-rank part. Across 10 seeds this locates a boundary near 4% tail energy, above which Muon’s equal-rate schedule (Kang et al., 2026) stops helping and GD’s greedy schedule wins. This reconciles the exact recovery we see for Muon in §5 with the removal of the simplicity bias reported for the same optimizer by Dragutinović et al. (2026). A closed-form solution based on a decoupled two-timescale approximation

recovers a timing mechanism compatible with that boundary (Proposition 8.1). The same consideration applies to practical tasks: on two hyperspectral image-completion benchmarks at equivalent train loss, GD achieves a 43–44% better held-out error than Adam under higher underdetermination, at a much lower effective rank, and in all four seeds (§9; learning rates chosen throughout by a train-only criterion). Most importantly, our framework is constructive: the coordinate-wise clip on the injected flow velocity in our own previously published optimizer is by itself enough to break equivariance, quietly suppressing the very bias it was supposed to inject. By contrast, a clip based on a global norm restores equivariance of the injected velocity and achieves the best bias restoration among the Adam-type methods we test—though the hybrid becomes fully equivariant only once Adam’s diagonal preconditioner is taken to its shared-scalar end as well.

Scope. This work deliberately avoids advocating any particular optimizer as a new state-of-the-art, and the mechanism does not promise benchmark wins. The bias it encodes is mild, and we highlight the conditions under which it is beneficial in §10.1: it wins over tuned explicit regularization only where the optimal regularization is itself mild and aligned in direction, and it falls short, in ways the theory anticipates, on tasks that demand stronger regularization or the kind of per-coordinate adaptation Adam provides. Like Gunasekar et al. (2017) and Arora et al. (2019), what we provide is a mechanistic explanation for an observed phenomenon, stated as a testable claim and evaluated in settings where it is expected to help as well as where it is not.

2 RELATED WORK

Implicit bias of gradient methods on factored models. Gunasekar et al. (2017) conjectured, and proved for commuting measurement matrices, that gradient flow on symmetric factorizations $UU^\top$ from infinitesimal initialization converges to minimum-nuclear-norm interpolants of underdetermined matrix sensing. Arora et al. (2019) extended the phenomenon to deep matrix factorizations and argued for a dynamical (rather than purely norm-based) picture, sharpened by Razin & Cohen (2020) and by the greedy low-rank learning dynamics of Li et al. (2021). Much of this analysis leans on the quantity $U^\top U - V^\top V$, which gradient flow conserves (Arora et al., 2018; Du et al., 2018). That conservation is bought by the same symmetry we study here: a continuous symmetry of the loss hands the flow a conserved quantity, in the Noether sense made precise for learning dynamics by Kunin et al. (2021).

This entire line of work studies gradient descent and gradient flow. Our transfer theorem extends its pathwise conclusions to memoryless common-scalar flows under the original assumptions, and our experiments address a question this literature leaves open: what do the optimizers people actually deploy do to this bias?

Implicit bias of Adam and adaptive methods. Wilson et al. (2017) first exhibited adaptive methods finding worse-generalizing solutions on constructed and real problems. The modern theory of Adam’s implicit bias is largely margin-based on separable classification, following Soudry et al. (2018) for gradient descent: ℓ_∞ -norm-constrained characterizations of AdamW (Loshchilov & Hutter, 2019) by Xie & Li (2024), Karush–Kuhn–Tucker convergence for Adam (Zhang et al., 2024; Cattaneo et al., 2024), per-sample refinements (Baek et al., 2026), and a unified steepest-descent view assigning Adam the ℓ_∞ and Muon the spectral geometry (Gronich & Vardi, 2026; Fan et al., 2025; Bernstein & Newhouse, 2024). Li et al. (2025) characterize a distinct sharpness measure that Adam minimizes near minimizer manifolds, driving it to qualitatively different solutions from SGD on overparameterized models. This is an adaptive-vs-SGD selection gap adjacent to ours, but it operates through a sharpness-reduction mechanism rather than a gauge mechanism, and it is not studied on factored-model recovery.

Xie et al. (2025) show that SGD’s optimization trajectory is rotationally invariant while Adam’s is basis-dependent, attributing Adam’s empirical efficacy to its alignment with coordinate (ℓ_∞) geometry. We examine the complementary aspect of this basis dependence, specifically isolating the selection cost it imposes. §10.1 details the conditions under which these trade-offs favor one algorithm over the other. To the best of our knowledge, none of these works explore low-rank recovery or interpolant choice in factored models, the setting where Adam’s ℓ_∞ -based coordinate structure interacts with Gunasekar-style implicit bias. That setting is the focus of this paper. DePavia et al. (2025) show that

small rotations of data or feature space alter the richness bias of Adam in classification settings, and propose an orthogonal reparameterization to alleviate this. Our work is complementary to theirs: we study the internal gauge of parameter factorizations and its impact on solution selection, and we characterize the optimizer zoo—the nine deployed update rules of §5—in relation to this property.

Optimizer design, symmetry, and structured methods. Table 1 summarizes the most closely related prior and concurrent studies, categorized by the symmetry examined and their relation to this paper. Appendix E covers this literature in full: transformation invariance in optimizer design, symmetry-driven accounts of implicit bias, and Muon and structured optimizers.

Table 1: Closest prior and concurrent work. The recurring distinction is whether an update respects a relevant orthogonal symmetry; we ask what it predicts for interpolant selection in factored models at matched training loss.

work	setting and result	relation to this paper
VectorAdam; Xie et al. (2025)	Examine ambient rotations of vector parameters under general loss functions. Show Adam is sensitive to such rotations, whereas shared scalar updates restore equivariance.	We focus instead on an internal factorization gauge and connect the same principle to solution selection among interpolants, rather than optimization speed.
LoRA-RITE; DePavia et al. (2025)	Study transformations of LoRA factors and rotations in data or feature space. Demonstrate improved optimization through invariance and altered implicit bias in Adam under rotations.	We isolate the intrinsic gauge of the factorization itself and evaluate recovery performance at matched training loss across a range of optimizers.
Silverstein et al. (2026); Zhang (2026)	Analyze per-head query–key (QK) rotations and Schur multiplicity bases. Highlight that optimizer design can intentionally break or preserve symmetry.	We maintain the gauge structure intact and use it to analyze solutions chosen by standard, deployed optimizers.
Lau & Su (2026); Shirodkar (2026)	Investigate architectural symmetry groups and construct optimizers that exactly respect equivariance.	These works support symmetry-aware design; we instead assess the practical consequences of preserving or violating such symmetry.
Kang et al. (2026); Dragutinović et al. (2026)	Study Muon’s spectral dynamics and its impact on simplicity bias.	We extend this by mapping conditions under which uniform spectral growth helps or harms performance as the target develops a spectral tail.

Our own prior work. FlowAdam ([Singh & Sheel, 2026](#)) enhanced Adam with clipped gradient-flow injection and demonstrated improvements on coupled problems using default settings. The current paper shifts focus from algorithm development to mechanistic explanation. In §10, we show that our symmetry criterion retrospectively clarifies FlowAdam’s design evolution: its original per-coordinate clipping introduced a gauge-breaking transformation that inadvertently constrained the very bias it aimed to introduce, whereas correcting this with global-norm clipping restores gradient descent’s bias within the hybrid method.

3 SETUP: GAUGE, TASK, AND PROTOCOL

Factored model and gauge transformation. Let $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ be a differentiable function, and consider the factored objective $L(U, V) = f(UV^\top)$, where $U, V \in \mathbb{R}^{n \times k}$ and the representation is overparameterized (k at least the rank of any solution of interest). For any orthogonal matrix $Q \in O(k)$, define the gauge transformation

$$\rho_Q(U, V) = (UQ, VQ).$$

Since $(UQ)(VQ)^\top = UV^\top$, it follows that $L \circ \rho_Q = L$: this gauge transformation is a symmetry of all such factored objectives, regardless of f . We consider the square case for notational simplicity; all

subsequent claims hold for rectangular inputs $f: \mathbb{R}^{d_1 \times d_2} \rightarrow \mathbb{R}$ with $L(U, V) = f(UV^\top)$, $U \in \mathbb{R}^{d_1 \times k}$ and $V \in \mathbb{R}^{d_2 \times k}$ (as in the hyperspectral experiments of §9), since the gauge only acts on the shared latent dimension k .

Definition 3.1 (Gauge equivariance of an optimizer). Consider an optimizer with internal state s_t (e.g. momentum terms, preconditioning matrices) and update rule

$$(U_{t+1}, V_{t+1}, s_{t+1}) = \mathcal{A}(U_t, V_t, s_t; \nabla L).$$

It is *gauge-equivariant* if for every $Q \in \text{O}(k)$ there is a corresponding transformation σ_Q on the state such that initializing from $(\rho_Q(U_0, V_0), \sigma_Q(s_0))$ leads to states $(\rho_Q(U_t, V_t), \sigma_Q(s_t))$ at every step t .

All methods we consider have zero-initialized states, and we impose $\sigma_Q(0) = 0$, ensuring that both trajectories start from the same initial conditions. That convention is not what excludes Adam: its entrywise second moment admits no state map σ_Q at all, whatever it does at zero, because squaring and right-rotation do not commute—which is exactly what the scalar version repairs (proof of Proposition 4.2(2)). For an equivariant optimizer it follows that the product matrices $W_t = U_t V_t^\top$ are invariant under the gauge action; that invariance is exactly what the coordinate-wise rules lose. This defines a trajectory on the orthogonal orbit of the initial pair (U_0, V_0) , which is not uniquely defined by W_0 alone (since the full symmetry group is $\text{GL}(k)$, and even gradient descent depends on the balance term $U_0^\top U_0 - V_0^\top V_0$, which can vary within a $\text{GL}(k)$ orbit). We restrict to the orthogonal subgroup $\text{O}(k)$, which is the largest subgroup acting by isometries (Lemma 3.2), under which gradients transform covariantly and Frobenius-norm statistics are preserved (see the attention-related footnote in §6).

Lemma 3.2 (Orthogonal gauge as maximal isometric symmetry). *For any $A \in \text{GL}(k)$, the transformation $(U, V) \mapsto (UA, VA^{-\top})$ preserves the product $UV^\top$. This transformation preserves the Euclidean product metric $\|U\|_F^2 + \|V\|_F^2$ for all (U, V) if and only if $A \in \text{O}(k)$.*

A critical property that makes this symmetry useful for verification is that gradients transform consistently with the factors:

$$\nabla_U L(UQ, VQ) = (\nabla_U L(U, V))Q,$$

and similarly for V (Lemma 4.1). Optimization updates constructed from operations that respect right multiplication by Q will therefore exhibit equivariance, while updates that apply a nonlinear operation entrywise in the (U, V) coordinates will generally not.

Task: recovery in the interpolation regime. Our test-case task is matrix sensing under the implicit-bias setup of Gunasekar et al. (2017) and Recht et al. (2010): sensing a planted matrix $X^* = U^* V^{*\top}$ ($n=40$) of rank $r^*=3$ through $m=2$ dof Gaussian measurements $y_i = \langle A_i, X^* \rangle$ with $\text{dof} = r^*(2n - r^*)$. We consider full overparameterization ($k=n$), small initialization (10^{-3}), no weight decay, and optimize the squared loss on the measurements. The parameter space is underdetermined: infinitely many matrices (low- and full-rank) are compatible with any given measurements.

All methods are stopped at interpolation (loss below 10^{-7} , final loss reported), to focus the comparison on differences in the set of possible minima reached once the model has enough expressivity to fit the data. Accordingly, all methods achieve interpolation (see Appendix D); but four of the nine considered—Muon, Lion, RMSProp, and signum (signSGD with momentum)—require learning-rate decay specifically shaped to their own dynamics (see below). We report recovery and effective rank,

$$\text{rec}(W) = \|W - X^*\|_F / \|X^*\|_F, \quad \text{erank}(W) = \exp\left(-\sum_i \pi_i \log \pi_i\right),$$

with $\pi_i = \sigma_i / \sum_j \sigma_j$ and $0 \log 0 := 0$ (Roy & Vetterli, 2007). Throughout, *recovery* means the error $\text{rec}(W)$, so **smaller is better** and “best recovery” always means the smallest value; the spectral shares π_i are unrelated to the dial parameter p of §7. Recovery is our main concern, as effective rank can be misleading: a method might reach a low-rank matrix in parameter space far from the true X^* (see §5).

Protocol. Implicit-bias experiments are notoriously rife with pitfalls and subtle interactions; some of the most critical ones are addressed by the following controls, which were applied in all experiments reported in the main text and the appendices. More discussion is in Appendix D.

- *No weight decay in any implicit-bias experiment.* Not only is implicit bias easily masked by explicit regularization, but interactions between the two (e.g. a regularizer placed in the loss versus decoupled from it) would require careful dissection to properly disentangle.
- *Run to full interpolation, or report inability to do so.* See Appendix D.7 for an example of a sign-descent method where early stopping (at loss 5×10^{-2}) creates the illusion of good recovery.
- *Symmetric scheduling.* Interpolation either with a fixed learning rate (preferred) or with cosine decay; methods with constant update norm (Muon, signum, Lion) are compared against schedules that decay the step size to zero, to avoid an easy win for methods that cannot interpolate at a constant step size. RMSProp needs decay for a different reason, given in Appendix D.7. This is a per-method criterion rather than a class-level one; it is informed by observations, rather than theory; and it is supplemented by all methods being compared with fully decayed schedules, so that differences in learning-rate decay cannot explain differences in outcomes (Appendix D.1).
- *Sweeps of learning rates, not single values.* Picking the best recovery value a posteriori is outcome bias; picking the biggest value compatible with convergence is speed bias. We therefore report full recovery-versus-learning-rate curves and frame discussion around asymmetries in achievable performance (Appendices D.2 and D.3), rather than points.
- *Report recovery and rank jointly, not as separate metrics;* use the same random seeds for all methods; and use deterministic, full-batch updates (where relevant) to compare related methods (§6).

4 THEORY: EQUIVARIANCE AS A CLASSIFIER OF UPDATE RULES

All proofs are provided in Appendix B. Once the orthogonal gauge is fixed, verifying covariance becomes straightforward. However, the implications go beyond algebra—they determine which update rules maintain the intrinsic geometry of the factorization, rather than depending on an arbitrary choice of basis.

Lemma 4.1 (Gradient covariance). *Let $L(U, V) = f(UV^\top)$ and let $Q \in O(k)$. Then*

$$\nabla_U L(UQ, VQ) = (\nabla_U L(U, V))Q, \quad \nabla_V L(UQ, VQ) = (\nabla_V L(U, V))Q.$$

Proposition 4.2 (Equivariant class). *The following update rules satisfy gauge equivariance as defined in Definition 3.1.*

1. **Gradient descent and Polyak/Nesterov momentum:** any linear combination of past gradients computed at appropriately transformed points.
2. **Scalar-preconditioned Adam:** a version of Adam where the per-coordinate second-moment estimate is replaced by a single scalar $\hat{v}_t$, obtained via bias-corrected exponential moving average (EMA) of gauge-invariant quantities (such as the average of all entries in $g^{\odot 2}$ across both factors). The update uses denominator $a_t = \sqrt{\hat{v}_t} + \epsilon$ and step direction $\Delta \propto \hat{m}_t / a_t$.
3. **Muon:** updates given by $\Delta = \text{msign}(M_t)$, where M_t is the momentum buffer and $\text{msign}(M) = AB^\top$ comes from the compact SVD $M = A\Sigma B^\top$ (with $\text{msign}(0) = 0$). Equivariance holds exactly under real arithmetic and remains valid for any finite-order Newton–Schulz approximation.
4. **Shampoo:** the step

$$\Delta = (L_t + \lambda I)^{-1/4} G_t (R_t + \lambda I)^{-1/4}, \quad L_t = \sum_{u \leq t} G_u G_u^\top, \quad R_t = \sum_{u \leq t} G_u^\top G_u,$$

with damping $\lambda > 0$, ensuring the matrix roots are well-defined even when matrices are rank-deficient early in training (our experiments use $\lambda = 1$; for $\lambda = 0$, interpret the roots on their active subspace). The subscripts L_t and R_t always refer to Shampoo’s left and right accumulators, never to the objective L . In particular, note that equivariance is achieved exactly in this setting even with damping.

In all cases, the resulting product of matrices $W_t = U_t V_t^\top$ is uniquely defined by the input data and the gauge orbit of the initial factors (U_0, V_0) , independently of the specific basis used for those factors. The results hold for the full-matrix cases described above, assuming real arithmetic. Implementations of practical interest that use variable splitting, mixed update rules, coordinate-wise clipping, or other non-equivariant operations on subsets of the parameters are a different matter: in these cases, equivariance is achieved only within the subsets of parameters updated by the stated rules.

Proposition 4.3 (Nonlinear coordinate-wise maps break the gauge). *Assume $k \geq 2$, and suppose that a memoryless update function Φ has the form $\Phi(G)_{ij} = \phi(G_{ij})$, where $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is a fixed function. If the update function satisfies*

$$\Phi(GQ) = \Phi(G)Q \quad \text{for all matrices } G \text{ and all orthogonal transformations } Q \in O(k),$$

then ϕ must be linear, $\phi(x) = cx$ for some real constant c . In other words, any optimization algorithm that applies a fixed nonlinear function at the level of coordinates, at the zero optimizer state, cannot be gauge-equivariant.

The reason is that by Definition 3.1, the zero state must be taken to itself by any gauge transformation. Both the original and the gauge-transformed run therefore start identically and apply the same update Φ , so equivariance requires $\Phi(GQ) = \Phi(G)Q$ for all G , which is only possible if ϕ is linear. Many standard optimization algorithms have such coordinate-wise nonlinearities in their first update—Adam, RMSProp, signSGD, Lion—while Adafactor carries them in its factored statistics and is analyzed separately in the proof. (Checking the equivariance condition for subsequent updates would require reasoning about how the optimizer state evolves under gauge transformations, but starting from the zero state avoids this complication.)

Proposition 4.4 (Adam has an exact first-step gauge defect). *Fix $\varepsilon > 0$ and consider bias-corrected Adam using the update convention $\theta^+ = \theta - \eta \hat{m} \oslash (\sqrt{\hat{v}} + \varepsilon)$, with elementwise operations, shared stepsize η , and initial moment estimates set to zero. Define the function*

$$D_\varepsilon(G) := G \oslash (|G| + \varepsilon),$$

applied entrywise, with $|\cdot|$ the entrywise absolute value. Let $G_U = \nabla_U L(U_0, V_0)$ and $G_V = \nabla_V L(U_0, V_0)$ denote the gradients at initialization, and let $(\tilde{U}_1, \tilde{V}_1)$ be the first iterate obtained from the gauge-rotated starting point $(U_0 Q, V_0 Q)$. Define

$$E_Q(G) := D_\varepsilon(GQ)Q^\top - D_\varepsilon(G), \quad D_U := D_\varepsilon(G_U), \quad D_V := D_\varepsilon(G_V).$$

Then the difference in the resulting factorized products after one step satisfies the identity

$$\begin{aligned} \tilde{W}_1 - W_1 = & -\eta [E_Q(G_U)V_0^\top + U_0 E_Q(G_V)^\top] \\ & + \eta^2 [E_Q(G_U)D_V^\top + D_U E_Q(G_V)^\top + E_Q(G_U)E_Q(G_V)^\top]. \end{aligned}$$

Equivalently, the second-order term can be rewritten as $\eta^2 [D_\varepsilon(G_U Q)D_\varepsilon(G_V Q)^\top - D_U D_V^\top]$.

This shows that Adam’s failure of gauge equivariance manifests already at the level of the represented matrix in the first step—not merely as a mismatch in internal states. For example, with $n = 1$, $k = 2$, $f(w) = \frac{1}{2}w^2$, $U_0 = V_0 = (1, 0)$, and rotation matrix $Q = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, the two resulting products after one step are $(1 - \eta/(1 + \varepsilon))^2$ and $(1 - \eta/(2^{-1/2} + \varepsilon))^2$, which differ whenever $0 < \eta < 2^{-1/2} + \varepsilon$.

So far the classification is a list of examples. For memoryless update rules it can be made exact, and the exact answer is narrow: equivariance forces the update to be a left preconditioner whose only input is the Gram matrix.

Theorem 4.5 (Structure: equivariant memoryless rules are Gram-determined left preconditioners). *Let Φ be a memoryless update rule defined on gradients $G \in \mathbb{R}^{n \times k}$ with full column rank ($k \leq n$). Then $\Phi(GQ) = \Phi(G)Q$ holds for all $Q \in O(k)$ if and only if*

$$\Phi(G) = H(GG^\top)G$$

for some matrix-valued function H depending only on the Gram matrix $GG^\top$ —a quantity invariant under gauge transformations. The function H can be canonically expressed as $H(GG^\top) = \Phi(G)G^+$, where G^+ is the Moore–Penrose pseudoinverse. In the square invertible case ($k = n$), this simplifies

uniquely to $H(P) = \Phi(P^{1/2})P^{-1/2}$. Full column rank is a generic property—an open, dense condition, holding with probability one under random initialization—so the rank-deficient set has measure zero and Φ is left unconstrained on it. The only point of that set the dynamics are forced through is $G = 0$, which occurs at (i) perfect interpolation or (ii) a degenerate stationary point at which one factor has become zero. That case carries no freedom either, since the update vanishes identically there: equivariance forces $\Phi(0) = \Phi(0)Q$ for any $Q \in O(k)$, so $\Phi(0) = 0$.

The algebra is elementary; the classification it buys is not. It demonstrates that the four cases in Proposition 4.2 are not ad hoc examples but the only shape available, up to the choice of representation. And it shows where the remaining flexibility sits. Right-gauge equivariance alone allows H to be any matrix-valued function of $GG^\top$. For instance, a fixed nonscalar left multiplier $\Phi(G) = DG$ preserves gauge symmetry without acting solely on individual singular values; even imposing left-orthogonal equivariance still permits reweighting in each singular direction to depend on the full spectrum (e.g. $\Phi(G) = \|G\|_F G$ is bi-orthogonally equivariant).

When H is further restricted to a fixed scalar function of the spectrum of $GG^\top$, Proposition A.1 reduces it to a single univariate *spectral transfer function* $h(\sigma)$: the update moves along the gradient’s own singular directions and rescales each one, with $h(\sigma) = \eta\sigma$ for GD and $h \equiv \eta$ for the exact polar map. That one coordinate is the schedule axis of §8. Stateful methods instead require explicit handling of their evolving covariant state. Appendix A states the spectral proposition, the extensions to stateful rules and to deeper factorizations, and what balancedness does inside the class.

Theorem 4.6 (Transfer theorem). *Let $a(t) > 0$ be a measurable function such that $1/a$ is locally integrable, and suppose $a(t)$ is determined solely by gauge-invariant statistics of the trajectory up to time t . Consider the scalar-preconditioned dynamics*

$$\dot{\theta} = -\nabla L(\theta)/a(t), \quad \theta = (U, V),$$

where θ is locally absolutely continuous, and define the time transformation $\tau(t) = \int_0^t du/a(u)$. Then the reparameterized trajectory $\tilde{\theta}(\tau) := \theta(t(\tau))$ satisfies

$$\tilde{\theta}'(\tau) = -\nabla L(\tilde{\theta}(\tau))$$

for almost every τ , meaning both trajectories follow the same path in parameter space. If in addition $\int_0^\infty du/a(u) = \infty$ —which holds, for instance, when a is bounded above—the transformed time reaches infinity, ensuring that if the gradient flow converges, so does the preconditioned version, to the same limit.

Consequently, any conclusion regarding a gradient flow which relies solely on the path or limit point is valid under the joint assumptions of the above theorem and the original result; this is true for the restricted-regime results (Gunasekar et al., 2017; Arora et al., 2019; Li et al., 2021). However, conclusions which are time-dependent, such as rates of convergence or hitting times, cannot be transferred because of the time rescaling. In the absence of the divergence condition, only the finite part of the path up to $\tau_{\max} = \int_0^\infty du/a(u)$ is shared.

Remark 4.7 (Scope of the transfer, and what equivariance does not buy). Theorem 4.6 applies to the memoryless, common-scalar continuous flow, so it reaches gradient descent and common-scalar-preconditioned GD and no further. Scalar-Adam carries a first-moment exponential moving average (EMA) and is therefore not strictly an instance of it; we treat its agreement with gradient flow as empirically observed rather than formally implied (Remark A.5, §5). Equivariance is what makes the transfer available, but on its own it is neither necessary nor sufficient for low-rank recovery. The second factor is the spectral schedule—the greedy, sequential growth of singular values from small initialization (Li et al., 2021). ScaledGD (Tong et al., 2021) is the clean counterexample on one side: gauge-equivariant, yet it equalizes the convergence rate across all the singular values and gives up the greedy behavior we rely on (erank ≈ 13.7 on the zoo task, not tabulated).¹ Appendix D.10 supplies the counterexample on the other side, a non-equivariant rule (long-anneal signum) that recovers anyway. Within the class, how hard a method flattens the spectrum (empirically GD < Shampoo < Muon on our runs) is the axis that remains; for memoryless updates it is the h of Proposition A.1, and §8 maps it.

¹Our implementation of ScaledGD uses a factor-aware preconditioner inspired by Tong et al. (2021), but serves here as an equivariant control with equalized schedule, since we do not have a tuned reproducibility baseline for ScaledGD.

5 THE OPTIMIZER ZOO

Design. We evaluate nine optimization algorithms on the sensing task described in §3. Four of these are theoretically expected to be equivariant: gradient descent (GD), scalar-Adam (using the $p=0$ setting from §7 with a gauge-invariant root-mean-square (RMS) scalar), Muon (Jordan et al., 2024), and Shampoo (Gupta et al., 2018). The remaining five are predicted to break gauge symmetry: Adam (Kingma & Ba, 2015), RMSProp (Tieleman & Hinton, 2012), signSGD with momentum (“signum”; Bernstein et al., 2018), Lion (Chen et al., 2023), and Adafactor (Shazeer & Stern, 2018). Experiments use three paired seeds, with recovery averaged across seeds. For each such method, we pick the learning rate yielding the best recovery among those on that method’s own grid that reach interpolation. This is selection on the outcome, but it is applied identically to every method and it favors the coordinate-wise ones, so the class gap it produces is a lower bound. Appendix D.2 replaces the single selected number with the full learning-rate curves and the “can/cannot” reading of them. The four methods labeled (*cosine*) in Table 2 require learning-rate decay to achieve interpolation, as discussed in §3: Muon, signum, and Lion because of their constant update norm, and RMSProp because of its ε -bounded denominator. All nine optimizers achieve the 10^{-7} threshold for interpolation (the loss is evaluated every 200 steps, so the faster methods are recorded several orders of magnitude below the threshold; for instance, Adam reaches a final loss of 1.2×10^{-11}). To put these numbers in perspective, we also report the solution to the convex problem suggested by gradient-flow theory: the minimum of the nuclear norm $\min \|X\|_*$ subject to $\langle A_i, X \rangle = y_i$, on the same three problem instances (experiments/nuclear_norm_reference.py). This forms the reference row of Table 2; the solve attains a feasibility residual of 7×10^{-16} and recovers X^* on two of the three seeds, while on the third it finds a solution of *strictly lower* nuclear norm than X^* (38.416 vs. 38.509), so X^* is not the minimum-nuclear-norm solution at $m = 2$ dof for that seed’s measurements. Main results are displayed in Table 2 and Figure 1; further control experiments over the scheduling choices and selection criteria are deferred to Appendix D.

Table 2: The optimizer-zoo map. Matrix sensing, 40×40 , rank 3 ground truth, $m = 2 \times \text{dof}$, no weight decay, three paired random seeds, all nine optimizer runs continued until interpolation (final training loss reported). Recovery is measured as $\|W - X^*\|_F / \|X^*\|_F$ (lower is better); erank is effective rank; bal is $\|U^\top U - V^\top V\|_F$; the notes column names the theoretical framework each row falls under—Gunasekar et al. (2017) for gradient descent (GD) and Wilson et al. (2017) for Adam. (*cosine*) marks the four methods that need cosine annealing to interpolate; the first row is a convex baseline, not an optimizer, recovering the minimum-nuclear-norm solution of the same measurements (both are unpacked in the text above). All values are means over the three seeds, with no dispersion quoted.

	method	recovery ↓	erank	bal	train loss	notes
	min-nuclear-norm	0.0335	3.30	—	—	convex reference, not an optimizer
equivariant	Muon (<i>cosine</i>)	0.0000	2.95	1.65	5.7×10^{-8}	near-exact (6.8×10^{-6} unrounded) Gunasekar anchor equivariant; flow proxy
	GD	0.1312	4.51	0.06	6.5×10^{-8}	
	scalar-Adam ($p=0$)	0.2010	5.43	0.14	3.5×10^{-8}	
	Shampoo	0.2856	6.95	1.47	4.5×10^{-8}	
coord.-wise	Lion (<i>cosine</i>)	0.4248	10.58	7.35	8.0×10^{-8}	low-rank-but-wrong needs decay; Appendix D.7 factored diag. still breaks Wilson anchor
	signum (<i>cosine</i>)	0.4454	7.83	1.79	6.5×10^{-8}	
	RMSProp (<i>cosine</i>)	0.5266	12.80	4.74	6.0×10^{-8}	
	Adafactor	0.5430	10.72	3.40	5.6×10^{-8}	
	Adam	0.5734	14.37	5.37	1.2×10^{-11}	

Result. The two clusters of predicted methods are cleanly separated: all equivariant methods perform strictly better than any coordinate-wise method (≤ 0.286 vs. ≥ 0.42), leaving a 0.14 gap between the best performer in the latter group and the worst performer in the former at this computational budget (see Figure 1). Because the table quotes three-seed means with no dispersion, the separation claim itself is deferred to the 10-seed ladder of Appendix D.9, where it persists at every problem size up to $n=256$ (GD, scalar-Adam, and Muon against the clean coordinate-wise methods, with two flagged exceptions). Since all nine methods manage to interpolate, the difference must be attributed to the choice of interpolant. The reference row lets us ground this in the absolute recovery error: compared to the min-nuclear-norm solution (0.0335), Muon is better, while GD,

scalar-Adam, and Shampoo are within an order of magnitude and the coordinate-wise approaches are all worse by more than an order of magnitude. Two points about that row are worth emphasizing. First, GD is worse than the convex benchmark and converges toward it as the step size decreases ($0.131 \rightarrow 0.113$, and 0.108 on the finer ladder grid) but never reaches it within this budget. This illustrates the distinction made in Remark A.5 between gradient *flow* at infinitesimal initialization and the discrete GD we actually run, at initialization scale 10^{-3} and on a finite budget: the nuclear-norm characterization is a restricted-regime statement about the former, while practitioners must contend with the latter. Second, while Muon outperforms the convex benchmark, this is not a contradiction, since on the single seed where nuclear-norm minimization did not return X^* , Muon did recover it; hence these methods do more than just approximate the convex relaxation. Muon is behaving differently because of its equal-rate schedule (Kang et al., 2026), a bias not related to any norm—the reading §8 develops.

We note three observations in particular. (i) Muon is capable of close to exact recovery (recovery error 6.8×10^{-6} , effective rank 2.95 instead of the true rank 3) on this task once cosine decay permits interpolation. This demonstrates a stronger inductive bias towards low-rank solutions than gradient descent shows at practical learning rates, as one would expect from its equal-rate spectral dynamics (Kang et al., 2026) when the true matrix is exactly low-rank. (ii) Adafactor eliminates the low-rank bias as completely as Adam does; the per-coordinate anisotropy of the factored diagonal approximation makes this a matter of gauge dependence rather than of memory footprint. (iii) Finally, signum showcases why recovery, and not effective rank, is the metric to rely on: with the lowest effective rank of any coordinate-wise method (7.8, against Shampoo’s 6.95) but a recovery error of 0.445, signum still falls among the worst optimizers for this task. Its updates are concentrated along only a few directions, and those directions are rarely aligned with the true solution. (When doubling the length of the annealing phase, signum is in fact the only optimizer to return to the low-recovery regime, but through a different and more volatile mechanism than the flowing optimizers, rather than by preserving the flow-induced bias. This is discussed further in Appendix D.10.)

Controls (Appendix D). The observed splitting is not due to learning-rate scheduling: under a single shared cosine decay we still get 9/9 on classification, and giving GD and Adam the same benefit of the doubt as Muon—the full 2×10^4 -step cosine decay horizon, no early stopping—does not improve their results (Table 6). The same is true for initialization scales of 10^{-3} and 3×10^{-3} , though it is less pronounced at 10^{-2} , where every method but Muon has lost the small-initialization edge (Appendix D.4). We also observe the same phenomenon in a larger setting (60×60 , rank 5), not tabulated: 0.000 (Muon), 0.077 (GD), 0.465 (Adam); see `experiments/zoo_size_check.py`. The only optimizer for which scheduling makes a difference is RMSProp: because of the ε -bounded denominator in its update, it cannot reach the interpolation threshold at a fixed step size, and it recovers 0.527 either way (Appendix D.7).

6 THE SAME GAUGE LIVES IN ATTENTION HEADS

Attention has the same symmetry. In an attention head with queries $q = W_Q x$ and keys $\kappa = W_K x$ (Vaswani et al., 2017) (written κ here, since k is the gauge dimension throughout), the logits are invariant to the choice of basis in the head dimension: they only depend on the weights through $q^\top \kappa = x^\top W_Q^\top W_K x$. For any orthogonal matrix A_h specific to the head, replacing (W_Q, W_K) with $(A_h W_Q, A_h W_K)$ leaves all logits unchanged, hence all outputs of the model. In other words, picking a basis in the head dimension is a gauge freedom, and $W_Q^\top W_K$ plays the role of $UV^\top$ (under the convention $U = W_Q^\top$, $V = W_K^\top$).² Hence all statements from §4 carry over to individual attention heads.

This symmetry has been previously noted and deliberately broken: in particular, Silverstein et al. (2026) induce a preferred direction by adding an unlearned bias to q and v , resampled at random each batch, in the spirit of a Hamiltonian/Noether framework for memory-efficient optimizers. We instead retain the gauge symmetry and study the interplay between standard optimizers and it.

²The full function-preserving group of a head’s logits is larger (any invertible A acts by $(W_Q, W_K) \mapsto (AW_Q, A^{-\top} W_K)$); as in Definition 3.1 we work with its orthogonal subgroup, the maximal part acting by *isometries* of parameter space (Lemma 3.2).

Table 3: Adam is basis-dependent in attention, and the noise twin separates structure from chaos. The values are relative logit distances between twins on validation inputs for the same task; multi-seed statistics are shown in Appendix D.5. At the harness’s default CPU precision, the baseline rounding threshold is the step-0 column ($\approx 2 \times 10^{-7}$, i.e. one function evaluated across two bases); the equivariant methods reach this floor at step 1, and their later nonzero entries result from numerical noise, not from gauge-symmetry violations—Muon’s split in particular, as per Proposition 4.2. These discrepancies are reduced, if not eliminated, in GPU float64, where the step-1 values fall within 2.8×10^{-16} – 1.1×10^{-15} over the three methods and four configurations; Table 9 carries the worst case in each.

optimizer	twin type	step 0	step 1	step 100	step 1500 (final)
Adam	gauge	1.8×10^{-7}	3.6×10^{-3}	6.5×10^{-1}	7.7×10^{-1}
Adam	$A=I$	0	0	0	0
Adam	noise (10^{-7})	2.6×10^{-7}	2.9×10^{-7}	1.7×10^{-5}	1.6×10^{-5}
SGD	gauge	1.8×10^{-7}	2.9×10^{-7}	3.7×10^{-5}	2.0×10^{-5}
scalar-Adam	gauge	1.8×10^{-7}	1.6×10^{-7}	4.1×10^{-6}	4.3×10^{-6}
Muon	gauge	1.8×10^{-7}	2.2×10^{-7}	2.3×10^{-2}	8.5×10^{-1}
Muon	noise (10^{-7})	2.6×10^{-7}	2.9×10^{-7}	4.0×10^{-2}	8.7×10^{-1}

For a given head, define $U = W_Q^\top$, $V = W_K^\top$, and $Q = A_h^\top$ (this Q is the gauge element of §3; the Q in W_Q is the query label, and the two are unrelated). Then, by Proposition 4.4, the exact first-step deviation in the invariant $W_Q^\top W_K$ under Adam at zero optimizer state is given by the expression in that proposition. The subsequent drift in logits is then an empirical consequence of this deviation at the product level, which the twin protocol isolates.

Twin protocol. We train two 2-layer 4-head transformers with embedding dimension $d=64$ on modular addition modulo 47 (the arithmetic task of Power et al., 2022), using full-batch training on deterministic CPU, with no bias terms or qk-norm (to retain the gauge symmetry). In the second model, we replace each head’s (W_Q, W_K) by $(A_h W_Q, A_h W_K)$ at initialization, with A_h a random orthogonal matrix drawn per head. As both models realize the same function at initialization, any deviation during training is due to the optimizer being sensitive to the internal basis. We also have a twin with $A=I$ (the identity matrix), which should be bit-identical and serve as a sanity check for reproducibility; a noise twin with the same basis but with W_Q and W_K perturbed by a Gaussian of magnitude 10^{-7} , probing the chaos amplification due to small functional differences; and runs with equivariant optimizers. Of the latter, SGD is applied with heavy-ball momentum 0.9, and Muon follows the standard hybrid procedure: it updates the matrix parameters—where W_Q , W_K , and hence the head gauge live—while Adam optimizes the embeddings and the output head. The gauge acts only on the parameters Muon updates, and the remaining gradients are identical across twins, so the hybrid is equivariant for the symmetry under test (see Appendix C).

Results (Table 3, Figure 2). The Adam twins diverge right away: after one step, their logit difference is already 3.6×10^{-3} , as opposed to 2.9×10^{-7} for the noise twin at the same step (a difference of four orders of magnitude). The relative logit distance then saturates at 0.77. Meanwhile, the $W_Q^\top W_K$ invariants have a relative Frobenius distance of 56% (QK drift: mean relative Frobenius distance of per-head invariants, see Appendix C; not shown in Table 3, which reports logit distances), which manifests as a substantial shift in the gauge-invariant structure of the model, beyond mere rotations of individual heads. That last number is the one with a downstream consequence, which we take up at the end of this section.

The two perturbations are not directly comparable, as they act on different scales: the gauge transformation leaves the function of the model intact, but changes the parameters by $O(1)$. The logit differences at step 0 for the two twins are 1.8×10^{-7} vs. 2.6×10^{-7} (Table 3; the gauge twin’s step-0 value is rounding, one function evaluated in two bases, which is why the same-basis $A=I$ twin is exactly 0), so the noise twin serves as a probe of how sensitive the optimizer is to a function difference of that magnitude when no basis has changed. *In short, what Adam learns in an attention head depends on an arbitrary choice of coordinates that has no effect on the model’s input–output behavior.*

SGD and scalar-Adam twins sit at the harness’s numerical-noise level at step 1 and stay four to five orders of magnitude below Adam’s structural split thereafter, so replacing Adam’s per-

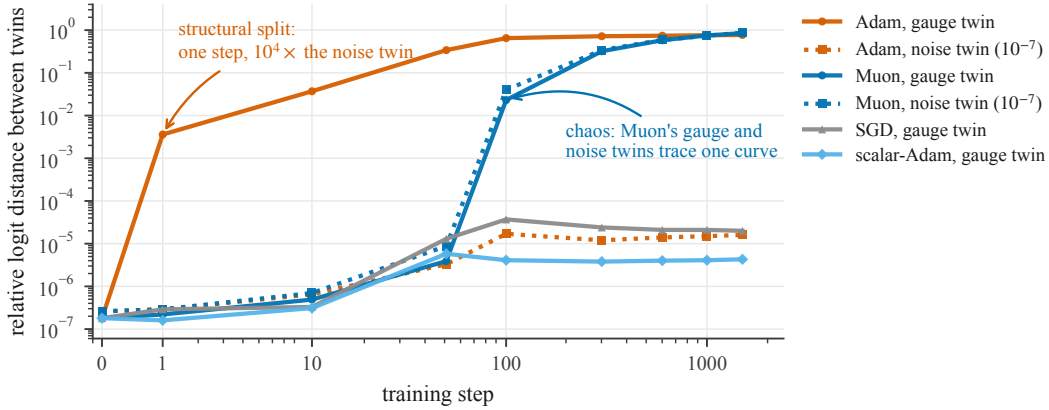


Figure 2: Adam is basis-dependent in attention, shown here by the relative logit distance between twins in the gauge and noise experiments. In the former the two copies are initialized as the same function, while in the latter they differ by a small (10^{-7}) perturbation in the same basis. Adam’s gauge twins (solid vermilion line) split structurally in one step, to a value four orders of magnitude greater than its own noise twin (dotted line), and then saturate. The same is not observed for SGD (heavy-ball momentum 0.9) or scalar-Adam, whose gauge twins remain at float noise. For Muon the gauge and noise twins trace the same curve, indicating that its separation is numerical chaos stemming from msign, not basis dependence.

coordinate denominator by a shared scalar—the $p=0$ end of the dial we build next (§7)—kills the basis dependence here as well as on the sensing task. Muon’s gauge twin also diverges, but its noise twin tracks the same trajectory up to a certain point (Appendix D.5): the agreement is within 0.05 decades averaged over the trajectory past step 100 (at step 100 itself the two still differ by 0.24 decades, Table 3), and both twins sit at the noise floor for about 50 steps, before numerical chaos takes over due to the instability of msign, not basis dependence.

Proposition A.1 shows that any spectrum-flattening schedule massively amplifies small singular values: unboundedly for the exact msign, and in practice by a large but finite Newton–Schulz factor around zero, which is where msign acts in early training with near-zero momentum. To disentangle these effects, note that the noise twin separates chaotic amplification by the schedule from genuine basis dependence during training. We thus propose using the triplet (gauge twin, $A=I$ twin, noise twin) as a diagnostic for claims of equivariance during training.

The phenomena are reliably observed across different random seeds and initialization draws: six (init $\times$ gauge-draw) pairs for Adam and two for each equivariant method, with the noise twin run out to a $100\times$ stronger perturbation, past which Adam’s gauge split still stands clear of it by over $600\times$ (Appendix D.5).

Scale, and real stochastic training. These phenomena are not specific to toy models or CPU precision. They occur in both full-batch and stochastic settings, and are not specific to model depth (see Appendix D.5 for the same analysis in a GPU float64 setting). When rerun in float64 on GPU (deterministic, with $A=I$ twins exactly zero), for modular addition over the integers mod 97, Adam’s gauge twins diverge on step 1, by $8.2\text{--}11 \times 10^{-3}$ (4 layer/ $d=128$) and $6.5\text{--}7.2 \times 10^{-3}$ (6 layer/ $d=256$), for six init $\times$ draw pairs each, saturating at 0.61–0.75 gauge drift. By contrast, all the equivariant methods are confined to $2.8 \times 10^{-16}\text{--}1.1 \times 10^{-15}$ at step 1: about machine epsilon in float64, 12–13 orders of magnitude below Adam.

Similar findings obtain for the stochastic setting: for a 6-layer, $d=256$ character-level language model (tiny-Shakespeare, Karpathy, 2015), with all methods receiving identical minibatch streams, Adam’s gauge twins split at step 1 ($2.2\text{--}2.4 \times 10^{-3}$), while the equivariant twins stay at $3\text{--}8 \times 10^{-16}$. Critically, while the two Adam runs have very similar validation losses (1.579 vs. 1.585), they have very different functions; their logit distributions end up 0.36–0.37 apart in relative distance, depending on the gauge pair (Table 9).

At larger depths, the final gauge drift stops being an informative summary, as most methods’ twins separate by the end; SGD’s are the exception and stay at float level throughout, though at 6 layer/ $d=256$ SGD never leaves chance accuracy, so its zero drift reports an untrained model rather than a preserved

gauge (Appendix D.5). These observations are not specific to stochastic training: a similar effect is seen in the deterministic full-batch mod-97 experiments at 6 layer/ $d=256$, where even the equivariant twins start to separate, and it again manifests in the character-level LM. Wherever a method’s twins do separate, each one lands where its own noise twin lands (Appendix D.5 tabulates both), suggesting that late-time gauge divergence occurs due to Lyapunov instability amplified onto the gauge symmetry via finite precision arithmetic; the equivariance failure modes are subdominant to the amplified floating-point error. For Muon, the instability is the Newton–Schulz iteration’s sensitivity to small singular values, while for scalar-Adam it is limited precision arithmetic (the depth and minibatching acting as additional amplifiers). Only Adam moved at step 1 (a change $39\times$ larger than even the $100\times$ stronger noise twin), while the equivariant methods were confined to float precision. In this picture, early basis-dependence is the discriminating signal, and this trend appears to generally hold (Appendix D.5, Table 9).

A consequence for model merging. Aligning two networks’ internal bases before averaging them (permutation re-basin, Ainsworth et al., 2023, and its rotational analogue here) presupposes that alignment makes the networks agree. The twin experiments make this premise optimizer-dependent inside every attention head. Alignment can at most reconcile the gauge, so two runs are head-mergeable only if the gauge-invariant product $W_Q^\top W_K$ agrees.

In the shallow deterministic regime, where the dynamics stay non-chaotic, our equivariant twins agree in the invariant to floating-point error and are mergeable by one Procrustes step; at 6 layers they separate as well, but only as far as their own noise twins do (Appendix D.5). Adam’s twins instead disagree in the invariant itself by 56%, so no per-head rotation can reconcile them, even though the runs began as the same function. Basis-dependent training does not simply pick among equivalent solutions; it removes the agreement that alignment-based merging needs. We have not tested merging end to end, so this is a consequence of the measured invariant rather than a merging experiment.

7 ISOLATING PRECONDITIONER ANISOTROPY WITH A DIAL

The zoo sorts the optimizers, and the attention twins show the same sorting inside a transformer. Neither says which ingredient of Adam does the damage. Here, we employ a one-parameter family to isolate the specific factor driving these differences. We define Adam- p by modifying Adam’s denominator $\sqrt{\hat{v}} + \epsilon$ to $(\sqrt{\hat{v}} + \epsilon)^p (\bar{s} + \epsilon)^{1-p}$, where $\bar{s} = \sqrt{\text{mean}(\hat{v})}$ represents a shared root-mean-square scalar, and $\epsilon = 10^{-8}$ is the stabilizing term used in Adam. The choice of $\bar{s}$ is significant. Computing the arithmetic mean of $\hat{v}$ across the entries in both factors gives a gauge-invariant quantity, since the Frobenius-norm statistic to which it is equivalent is preserved under the right-orthogonal transformation of Lemma 4.1. As such, when $p=0$ the denominator reduces to the gauge-invariant scalar of Proposition 4.2(2). Adam-0 then coincides with the scalar-Adam procedure of §5 up to an additive $O(\epsilon)$ term in the denominator: it is the same algorithm, and the results are identical (0.201, erank 5.43). In contrast, the geometric mean of $\sqrt{\hat{v}}$ would fail to be gauge-invariant, rendering the $p=0$ case itself dependent on an arbitrary choice of factorization basis. We adhere to the RMS convention throughout, giving the geometric-mean alternative only as a consistency check.³

When $p=1$, we recover standard Adam, which is coordinate-wise and gauge-breaking. For intermediate values of $p \in (0, 1)$, the preconditioner is less anisotropic while retaining the same adaptation, momentum, and noise properties as standard Adam. The overall size of the denominator is controlled by p , so for the purposes of the experimental sweeps it makes sense to test either a shared set of learning rates for all p or a separately optimized learning rate for each p (see Algorithm 1 for the exact procedure, including the two places in which ϵ is used).

The resulting behavior on the zoo task is shown below, with improvements in both parameter recovery and effective rank as p decreases towards $p=0$, without any change to the algorithm itself, simply by varying p in the update rule. Two experimental setups were used to illustrate this result: a fixed-step version, where all learning rates were tested for all values of p with the other hyperparameters fixed as well, and an envelope version, where for each p the best learning rate was chosen individually based

³The geometric-mean dial traces the same monotone curve and agrees with the RMS dial at the endpoint to within 0.03 (recovery 0.229, erank 5.8 at $p=0$). It is an approximate rather than an exact dial, because its $p=0$ limit is not equivariant.

Algorithm 1 Adam- p : the preconditioner-anisotropy dial (§7). Setting $p=1$ recovers standard Adam; $p=0$ gives the gauge-equivariant scalar-Adam from Proposition 4.2(2). Only the denominator moves with p : the gradient, moment estimates, bias corrections, step size, and (zero) weight decay are the same throughout. All operations on the two factors are elementwise: $\odot$, $\oslash$, $(\cdot)^{\odot q}$ denote elementwise multiplication/division/power. The only inter-factor cross-talk happens at Line 8, which is precisely what makes the update equivariant at $p=0$: by pooling $\hat{v}$ over all entries of the two factors, we make $\bar{s}$ depend only on $\|G_U\|_F^2 + \|G_V\|_F^2$, which is invariant under gauge transformations by Lemma 4.1. For $p < 1$ an extra ε sits outside the product as well, so at $p=0$ the denominator is $\bar{s} + 2\varepsilon$ rather than the ideal $\bar{s}$; with $\varepsilon = 10^{-8}$ this $O(\varepsilon)$ offset has no effect on any number reported here. At $p=1$ that extra ε is dropped and the denominator is stock Adam’s $\sqrt{\hat{v}_\theta} + \varepsilon$. The second-moment term v_θ is kept elementwise for all p , including $p=0$; but since taking the entry mean commutes with the exponential moving average, the pooled $\bar{s}^2$ equals the bias-corrected scalar v_t of Proposition 4.2(2) exactly, making the parameter trajectory the equivariant one. It is this pooled scalar, not the stored elementwise v_θ , that carries the state map for the purposes of Definition 3.1.

Require: factors U, V ; step size η ; $(\beta_1, \beta_2) = (0.9, 0.999)$; $\varepsilon = 10^{-8}$; dial $p \in [0, 1]$; weight decay 0

```

1:  $m_\theta \leftarrow 0, v_\theta \leftarrow 0$  for each factor  $\theta \in \{U, V\}$ 
2: for  $t = 1, 2, \dots$  do
3:   for each factor  $\theta \in \{U, V\}$  do
4:      $g_\theta \leftarrow \nabla_\theta L(U, V)$   $\triangleright L(U, V) = f(UV^\top)$ ; covariant:  $g_\theta \mapsto g_\theta Q$ 
5:      $m_\theta \leftarrow \beta_1 m_\theta + (1 - \beta_1) g_\theta, v_\theta \leftarrow \beta_2 v_\theta + (1 - \beta_2) g_\theta^{\odot 2}$ 
6:      $\hat{m}_\theta \leftarrow m_\theta / (1 - \beta_1^t), \hat{v}_\theta \leftarrow v_\theta / (1 - \beta_2^t)$ 
7:   end for
8:    $\bar{s} \leftarrow (\text{mean}(\hat{v}_U, \hat{v}_V))^{1/2}$   $\triangleright$  one mean, pooled over both factors: gauge-invariant
9:   for each factor  $\theta \in \{U, V\}$  do
10:     $D_\theta \leftarrow (\hat{v}_\theta^{\odot 1/2} + \varepsilon)^{\odot p} \odot (\bar{s} + \varepsilon)^{1-p}$   $\triangleright p=1$ : per-coordinate;  $p=0$ : one shared scalar
11:    if  $p < 1$  then
12:       $\theta \leftarrow \theta - \eta \hat{m}_\theta \odot (D_\theta + \varepsilon)$ 
13:    else
14:       $\theta \leftarrow \theta - \eta \hat{m}_\theta \odot D_\theta$   $\triangleright D_\theta = \hat{v}_\theta^{\odot 1/2} + \varepsilon$  here: stock Adam
15:    end if
16:  end for
17: end for

```

Denominator convention: at $p=0$ it is $\bar{s} + 2\varepsilon$ and at $p=1$ it is $\hat{v}_\theta^{\odot 1/2} + \varepsilon$; both are the ideal value up to $O(\varepsilon)$.

on recovery in the interpolating regime. In the fixed-step version, for every learning rate whose sweep both interpolates and resolves the flow limit (for the zoo task, the range from 10^{-3} to 3×10^{-2}), performance was averaged over three random seeds, and parameter recovery consistently improved with decreasing p : from 0.576 to 0.201 at 10^{-3} , from 0.570 to 0.256 at 3×10^{-3} , from 0.573 to 0.357 at 10^{-2} , and from 0.581 to 0.493 at 3×10^{-2} , with every run meeting the 10^{-7} interpolation threshold. Effective rank follows the same trend (Figure 3).⁴

In the envelope setup, for each p , the learning rate was selected which produced the best possible recovery in the interpolating regime, according to the standard procedure for the zoo task (Appendix D.2). The resulting envelope also shows monotonic improvement and matches the fixed-step results up to three decimal places at every $p \leq 0.75$ when the learning rate is 10^{-3} , indicating that the per- p learning rate selection does not introduce hidden effects. Figure 3 displays the envelope sweep and provides tabulated results. Its $p=1$ endpoint reads 0.570 where the zoo’s Adam row reads 0.5734, because the dial’s shared grid selects a different rate; Adam is flat across that range (Appendix C).

This refines the finding of Wilson et al. (2017) into a continuous, adjustable relationship: along this tuning axis, the coordinate-wise anisotropy of the preconditioner directly explains the erosion of factored-model bias. The same anisotropy is also what buys Adam’s speed—switching to $p=0$

⁴At the grid’s largest rate (10^{-1}) every p still interpolates, but recovery sits at 0.55–0.62 and effective rank at ≈ 13.8 – 15.3 for all p : a step this coarse leaves every endpoint far from its flow limit (Remark A.5), so the sweep flattens rather than restores. The depth of the restoration is rate-dependent; its monotonicity, at rates that resolve the flow limit, is not.

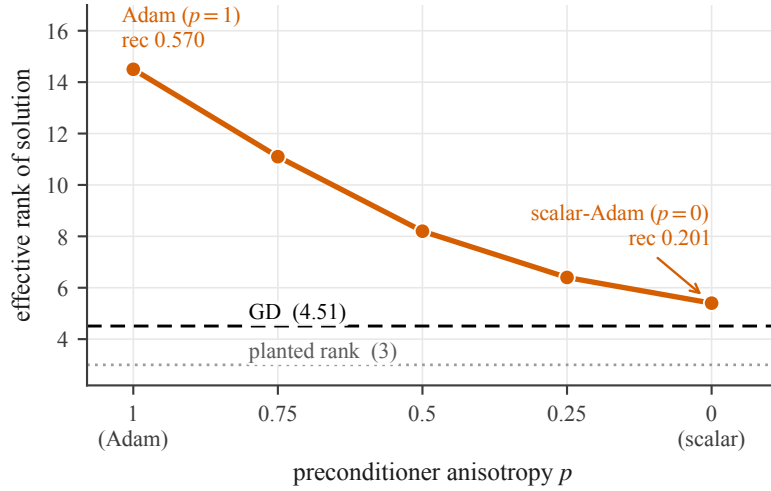


Figure 3: The dial. As Adam’s preconditioner changes shape from coordinate-wise ($p=1$) towards isotropic ($p=0$), the effective rank of the solution gracefully decreases towards that of standard gradient descent (GD). The plotted curve is the “envelope arm,” in which for each p we pick the learning rate giving the best recovery from a shared grid. A “fixed-step arm” that instead varies only p at each individual rate of that grid agrees with it closely (§7). Since the two arms differ precisely in whether the learning rate is free to absorb the change in denominator scale that comes with p , their agreement points to per-coordinate anisotropy as the driving force behind the effect.

increases the number of steps needed for interpolation by roughly a factor of eight (from 400 to 3200 on this task, comparing each end at its selected rate). The balancedness drift of Proposition A.4 falls with the dial, and the endpoints bracket it: the zoo’s Adam row carries $\|B_T\|_F = 5.37$ and its scalar-Adam row 0.14 (Table 2), which ties the empirical adjustment to the conserved quantity. The monotonic trend is not tied to a specific problem scale: on H100, the pattern holds at $n=128$ (recovery improves from 0.67 to 0.40, effective rank from 36.6 to 12.0 as p goes from 1 to 0, across five seeds) and replicates at $n=40$ with ten seeds (recovery improves from 0.52 to 0.14 on an extended flow-limit grid, both ends selected at 3×10^{-4} , under every rate the zoo grid carries, which is why the $p=1$ end sits below the 0.56–0.58 of Remark A.5; the $n=128$ endpoint reads 0.40 against the ladder’s 0.345 for the grid reason above).

8 INSIDE THE EQUIVARIANT CLASS: THE SPECTRAL-TAIL PHASE DIAGRAM

Equivariance is binary, but behavior inside the equivariant class is not—for memoryless spectral-separable maps, Proposition A.1 gives a coordinate: GD has $h(\sigma) = \eta \sigma$, whereas the exact polar map has $h \equiv \eta$. Recent work highlights two seemingly contradictory trends for Muon: a strong simplicity (spectral) bias (Fan et al., 2025; Gronich & Vardi, 2026), and, for deep-linear and linear-attention models, a loss of GD’s sequential simplicity bias that makes Muon prone to fitting spurious features (Dragutinović et al., 2026). Our sensing and real-data experiments show both. In this section we explain how the spectral tail of the target reconciles these trends.

Design. Plant $X = \sqrt{1 - \tau^2} X_3 + \tau E_\perp$ where X_3 is the rank-3 matrix and $E_\perp$ a dense rank- $(n-3)$ component supported on the orthogonal complements of X_3 ’s column and row spaces, with $\|X_3\|_F = \|E_\perp\|_F = 1$. The support condition gives $\langle X_3, E_\perp \rangle_F = 0$, hence $\|X\|_F = 1$, so that $\tau \in [0, 1]$ and τ^2 is exactly the fraction of target energy in the tail. Sense with $m = 2$ dof counted at the rank-3 target, as before. All methods interpolate, and we sweep τ . The $\tau=0$ target is the zoo task of Table 2, but this sweep is a separate 10-seed run on its own two-rate grid, so its $\tau=0$ column reproduces that table’s *ordering* rather than its *digits* (GD 0.112 here vs. 0.131 there, and so on).

Table 4: Spectral-tail phase diagram (recovery; 10 seeds, float64). Muon is exact at $\tau=0$, degrades fastest as tail energy grows, and cedes to GD in a crossover region near $\tau^* \approx 0.2$ ($\approx 4\%$ tail energy). Bold marks each row’s best recovery, with both members of a within-noise tie bolded; the seed dispersions behind those ties are the ± 1 s.d. bands of Figure 4, omitted here for space. Decay-symmetrized rows in Appendix D.

τ	GD	Adam	Muon	Shampoo	regime
0.00	0.112	0.542	0.000	0.334	Muon exact
0.05	0.150	0.543	0.095	0.343	Muon
0.10	0.214	0.555	0.191	0.375	Muon
0.20	0.351	0.597	0.354	0.449	boundary ($\approx 4\%$ tail); Muon cedes
0.35	0.549	0.685	0.580	0.576	GD
0.50	0.727	0.776	0.752	0.723	tail regime; GD/Shampoo within noise

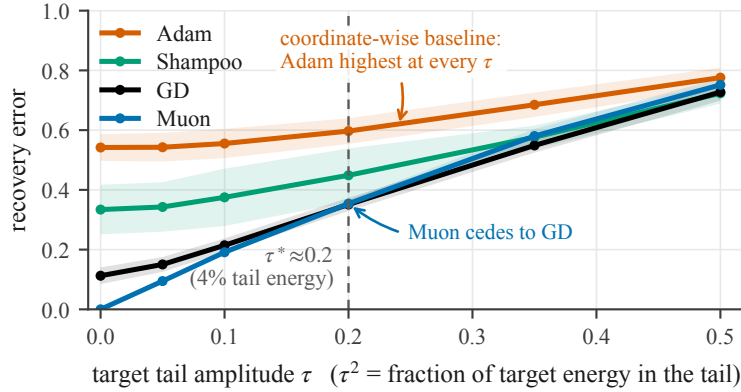


Figure 4: The spectral-tail phase diagram. Recovery vs. the target’s tail-amplitude parameter τ (the tail carries τ^2 of the target’s energy). Adam has the worst recovery error at every τ plotted. Within the equivariant class, Muon’s aggressive equal-rate schedule is exact at $\tau=0$ and crosses over to GD near $\tau^* \approx 0.2$. At the largest τ , all methods are near the measurement-determined floor set by the unidentifiable tail, and GD and Shampoo are within seed noise, so the ordering there is uninformative. Bands show ± 1 s.d. over 10 seeds. The boundary is unchanged when every method is given the identical cosine schedule (Appendix D).

The two axes in the sweep. Table 4 tabulates the sweep and Figure 4 plots it. On the symmetry axis, Adam has the highest recovery error at every τ in Table 4. Along the schedule dimension of the equivariant class, the degree of schedule aggressiveness (with Muon the most aggressive, followed by Shampoo and then GD) trades accurate low-rank recovery, which the aggressive schedules win, against robustness to the tail, which the moderate ones win. This creates a crossover region around 4% tail energy (the difference between GD’s and Muon’s performance at $\tau=0.2$ is within the range of the seed-to-seed variance). Thus, Muon’s preference for uniform growth across modes (Kang et al., 2026) is optimal if the signal is truly low-rank, but hurts performance on signals that carry a tail which should not be fit. Hence, there is a spectral regime in which each of the schedules is preferable, depending on the properties of the target signal.

This preference boundary does not depend on the type of decay schedule used: re-running the experiment with cosine-decayed schedules instead of constant ones keeps each method’s performance curve as a function of τ unchanged, with the crossings left where their constant-schedule counterparts put them. That control ran on the 3-seed grid, whose crossing is $\tau^* \approx 0.35$ under both schedules; the estimate then became more precise, shifting to approximately 0.2 on a 10-seed grid (see Appendix D.1).

An analytically tractable model. In the simplified framework of decoupled dynamics—commonly used in studies of greedy dynamics—it is possible to precisely characterize both sides of the trade-off and explain why the scale of initialization influences one learning schedule but not the other.

Proposition 8.1 (Greedy vs. equal-rate schedules: a solvable two-timescale scenario). *Consider the factored dynamic system decomposed into independent, balanced scalar components ($w = uv$, with $u = v = \sqrt{w}$; aligned target; this is the idealization adopted in Arora et al. (2019) and Li et al. (2021)). Suppose target modes satisfy $s_1 > s_2 > 0$, all initialized at $w_i(0) = w_0$ where $0 < w_0 \leq s_2/2$, and let*

T_1 denote the first time the leading mode reaches $(1 - \delta)s_1$, with δ fixed such that $0 < \delta < 1 - s_2/s_1$. (Both lower bounds are strict and necessary: $w_0 = 0$ is a stationary point, $\delta \leq 0$ would make the threshold unreachable, and the proof involves division by these terms.) (i) Greedy schedule ($h(\sigma) = \eta\sigma$): Each mode follows logistic growth, $\dot{w}_i = 2\eta w_i(s_i - w_i)$, and at time T_1 the smaller mode remains inactive:

$$w_2(T_1) \leq C_{\delta,\rho} s_1 \left(\frac{w_0}{s_1}\right)^{1-\rho} \xrightarrow{w_0 \rightarrow 0} 0, \quad \rho := s_2/s_1, \quad C_{\delta,\rho} = 2\left(\frac{1-\delta}{\delta}\right)^\rho.$$

Thus, reducing initialization enhances the timescale gap between leading and trailing modes indefinitely. (ii) Equal-rate schedule ($h \equiv \eta$): All modes grow at the same rate ($\sqrt{w_i}$ increases linearly with time) and saturate at their respective s_i ; by T_1 , the smaller mode is already fully fitted, $w_2(T_1) = s_2$, regardless of w_0 . Here, the flat schedule does not let the initialization scale differentiate between the modes.

Proposition 8.1 (proved in Appendix B) shows that the decoupled-mode model has a transient dynamics consistent with the observed phase behavior. In particular, if the target matrix has exactly low rank, uniformly fitting all the planted modes recovers the full signal in this model, as suggested by the equal-rate principle of Kang et al. (2026) for Muon. In the case of a target matrix with a less-determined tail (sensed with $m = 2$ dof measurements counted at the head), the flat schedule biases the fit towards directions the measurements do not constrain, while the greedy schedule suppresses those unconstrained modes down to the level of $w_0^{1-s_2/s_1}$. Thus, the observed transition may be determined by a combination of the spectral properties of the target matrix, the measurement design, the initialization scale, the training length, the schedule, and the optimization algorithm.

9 REAL DATA AT MATCHED TRAINING LOSS

While the synthetic data was constructed to have a known ground truth, real-world matrices are expected to have spectral tails. As discussed in §8, this suggests that the same mechanism should be at work in practice, with additional contributions from the tail components. We test this hypothesis on two hyperspectral image-completion benchmarks. The first is the Indian Pines dataset (Baumgardner et al., 2015), consisting of 2000 randomly selected pixels across 200 spectral bands, derived from a corrected Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) cube with water-absorption bands excluded. The second is Pavia University, featuring 2000 random pixels and 103 bands captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. In both cases, the spectral data are approximately low-rank due to physical properties, and both datasets were made publicly available by Graña et al. (2021).

We perform rank-48 factored matrix completion without weight decay, testing at two underdetermined sampling densities per dataset, calibrated so that $m/\text{dof}_{24} \approx 1.15$ and 1.9. Here, degrees of freedom (dof) are based on the intrinsic rank of 24—where each singular-value spectrum stabilizes—rather than the higher model rank of 48. This ensures the ratio reflects how underdetermined the true signal is (see Appendix C). The intrinsic rank is determined directly from each scene’s singular spectrum, sets a fixed sampling density for all methods, and is not tuned using validation performance. We refer to this ratio as m/dof henceforth.

A key challenge in comparing optimization methods is early stopping: slower algorithms may appear superior simply because they benefit from implicit regularization due to incomplete fitting. To address this, we compare methods along their full (training loss, test error, effective rank) trajectories at matched training loss levels. Specifically, for any given training loss ℓ , we record each method’s held-out root-mean-square error (RMSE) at the first point it reaches ℓ . Learning rates are selected using a training-only criterion (deepest convergence followed by fewest steps; see Appendix D.6), and results are averaged over four random seeds. That rule sets the size of every gap we report below. It selects for fast convergence within the budget, hence for larger rates, which puts GD in its rate-invariant region while handing Adam the worst rate on its own grid: giving each method the rate that minimizes its own held-out error would leave GD ahead by 13% rather than 43%. We keep the train-only rule because selecting and evaluating on the same held-out statistics would reintroduce the bias the protocol exists to avoid, and neither the ordering nor the rank profile changes under the other one (Appendix D.6).

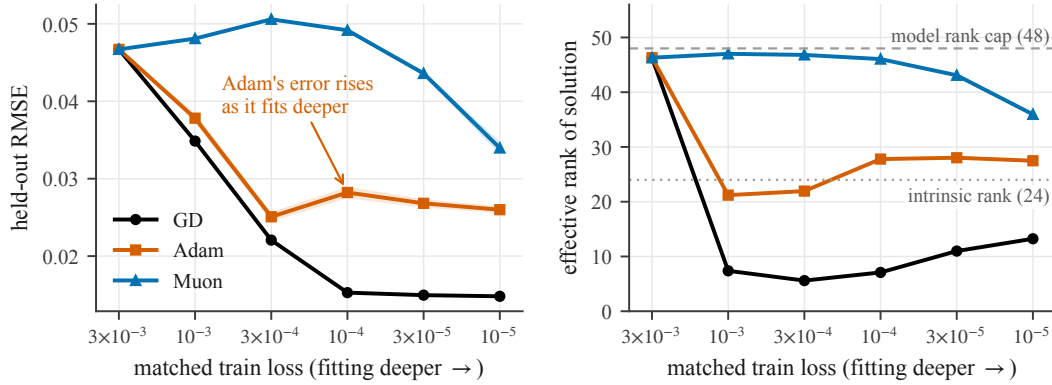


Figure 5: Performance as a function of training loss on real data (Indian Pines, $m/\text{dof} \approx 1.15$, 4 random seeds, GPU float64; learning rates chosen via the train-only rule in Appendix D.6). This is the corresponding row of Table 5, read along its whole path rather than at its endpoint alone. Gradient descent (GD) sits below Adam at every matched training loss beyond the shared starting point, and the two are compared only at equal training loss, so no fitted run is ever set against an unfitted one. *Left panel:* held-out error; bands are ± 1 s.d. over the 4 seeds, hidden inside the line width wherever the seeds agree—the whole path for GD—and opening up only where they do not, as Adam’s do once overfitting begins. *Right panel:* effective rank. GD settles between 5.6 and 13.2; Adam climbs to 28, against the scene’s intrinsic rank of 24; Muon tracks the rank cap until convergence, then drops sharply.

At matched training loss $\leq 3 \times 10^{-5}$, at an observed entry density of 0.15 ($m/\text{dof} \approx 1.15$), gradient descent (GD) achieves a held-out RMSE of 0.0150 ± 0.0001 at an effective rank of 11, while Adam reaches 0.0268 ± 0.0005 at rank 28. Thus GD’s held-out error is 44% lower than Adam’s at equivalent fit, in all four seeds, aligning with the rank behavior predicted by the mechanism.

The full learning trajectories offer more insight than final performance alone (Figure 5). GD’s test error decreases steadily as training progresses, consistent with a greedy, head-first fitting pattern. In contrast, Adam’s test error increases during interpolation (from 0.0251 at training loss 3×10^{-4} to 0.0268 at 3×10^{-5}), as its effective rank rises from 22 to 28. This shows that, on real data, deeper fitting with an optimizer that breaks gauge symmetry can actually degrade generalization. Muon shows little improvement in held-out error until very late stages, with effective rank remaining close to the model limit (dropping from 46 to 36)—a real-data manifestation of the tail behavior described in §8.

The performance gap narrows as sampling density increases, as predicted by the selection mechanism: the advantage of GD reduces from +44% at $m/\text{dof} = 1.15$ to +28% at $m/\text{dof} = 1.9$ (consistent across all four seeds), and appeared to fall to nearly zero by $m/\text{dof} \approx 3$ in an earlier interpolation-based evaluation, which we do not tabulate because the matched-loss protocol supersedes it. As more data become available, the solution becomes better constrained, reducing the impact of interpolant selection—a transition noted in §10.1.

Validation and a second benchmark. The observed effect is neither an artifact of where the comparison is taken nor specific to one dataset. Read at the stricter threshold ($\leq 10^{-5}$, the level Table 5 reports), the same Indian Pines runs give +43.0% at $m/\text{dof} \approx 1.15$ and +27.6% at 1.9, against the +44% and +28% quoted above at 3×10^{-5} : the gap narrows slightly as the fit deepens, and does not turn on the choice of stopping level.

A second independent hyperspectral dataset, Pavia University, matched to the same two density levels, exhibits a consistent trend: +43.8% improvement at $m/\text{dof} \approx 1.15$ and +22.9% at $m/\text{dof} \approx 1.9$ (results consistent across all 4 seeds; see Table 5). Muon does not outperform GD on either dataset or density, and its effective rank stays near the model limit (46–47 of a capacity of 48) for most of the matched-loss path, declining only in its final levels, earlier on Pavia than on Indian Pines—the tail regime of §8. On Indian Pines, Muon is actually worse than Adam, the opposite effect to the one observed in the synthetic experiments (Table 4), where Adam was significantly worse at every τ ; on Pavia, where that collapse does run to completion (rank 10 and 8 of 48), Muon lands back ahead of Adam at both densities. While being equivariant influences which mechanisms a method can access, it does not guarantee improvements in performance over the alternatives in all spectral settings.

Muon’s persistently high effective rank suggests that it is the schedule dimension that determines the behavior of a method here, not the gauge. Scalar-Adam is absent from the table because its rate grid here was undersampled and it does not reproduce the phenomenon (Appendix D.6). The GD–Adam gap carries the main matched-loss conclusion of this section (Table 5).

Table 5: Matched-loss recovery on two hyperspectral datasets (held-out RMSE $\times 10^{-2}$, mean $\pm$ standard deviation over 4 random seeds, evaluated at matched training loss $\leq 10^{-5}$, GPU float64; learning rates as in Appendix D.6). The reduction column reports the percent improvement of GD over Adam, as $(\text{Adam} - \text{GD})/\text{Adam}$ using mean performance over seeds; gradient descent (GD) outperforms Adam in every individual seed of every entry. In parentheses, Muon’s effective rank at the deepest point in the fit (after the collapse typical of the late stages), given a model capacity of 48.

dataset	m/dof_{24}	GD	Adam	RMSE reduction	Muon (rank)
Indian Pines	1.15	1.481 ± 0.013	2.600 ± 0.047	+43.0%	3.397 ± 0.082 (36)
	1.9	1.244 ± 0.012	1.718 ± 0.035	+27.6%	3.210 ± 0.122 (35)
Pavia Univ.	1.15	0.819 ± 0.026	1.458 ± 0.123	+43.8%	1.260 ± 0.121 (10)
	1.9	0.615 ± 0.010	0.798 ± 0.023	+22.9%	0.770 ± 0.017 (8)

10 REPAIRING OUR OWN OPTIMIZER USING THE CRITERION

FlowAdam (Singh & Sheel, 2026) introduces clipped gradient-flow velocity into Adam’s momentum mechanism, aiming to incorporate the dynamics responsible for bias as described in the memoryless, idealized form of Theorem 4.6—though the actual stateful hybrid version falls outside the theorem’s formal reach. Without weight decay, however, it only partially closes the performance gap to gradient descent (GD), reducing the recovery error from 0.573 to 0.347, compared to GD’s 0.131. The symmetry criterion exposes the flaw in the design of this algorithm: coordinate-wise velocity clipping, $v \mapsto \text{clip}(v, -c, c)$, acts independently on each coordinate and so breaks gauge symmetry, by Proposition 4.3. In short, the component designed to restore gauge-consistent behavior breaks it by the very act of being there.

By design, the clipping is the binding constraint: as its threshold $c \rightarrow \infty$ the clip switches off and the modified dynamics reduce to the underlying gradient flow. The appropriate way to modify the clipping is to restrict the whole velocity vector’s norm, not individual coordinates, which constrains magnitude without altering direction. It is this correction that brings FlowAdam’s bias to match the level of GD in the continuum limit (as $c \rightarrow \infty$ it recovers at 0.113, which is GD’s own value at its smallest interpolating rate in Table 7, versus the 0.131 of GD’s benchmark row). Applied alone, it improves the recovery error by 37% ($0.347 \rightarrow 0.220$, this run stopping at a training loss of 6×10^{-7} rather than the 10^{-7} threshold, so it is a deep fit and not an interpolating reading; Appendix C). The clip is all that this repairs, though: the surrounding Adam step still preconditions per coordinate, so the method as a whole remains gauge-breaking, and 0.220 is where that leaves it. Combined with the preconditioner-anisotropy dial from §7 (making it FlowAdam- p), it gives the best recovery among the Adam-type variants we test (0.169 versus Adam’s 0.573), with the flow contributing a further +15.9% over the dial alone (0.169 vs. 0.201) at the same learning rate. Furthermore, its effective rank (4.8–5.4 across different random seeds) is the closest among all the considered Adam-type algorithms to GD’s 4.51.

The +15.9% improvement is achieved at the same learning rate (both at 10^{-3} , as per the dial’s anchoring procedure in §7), and should not be viewed as an improvement in the asymptotic regime: on finer grids with smaller learning rates the two methods converge empirically, the dial alone reaching 0.143 ($n=40$) / 0.40 ($n=128$) while FlowAdam- p achieves nearly identical results at 0.148 / 0.42. That $n=40$ cell is the slowest to fit and clears the interpolation bar only past the sweep’s 3×10^4 steps; continued to a true 10^{-7} , all ten seeds cross by 6×10^4 and read the same 0.1475 ± 0.0357 . The evidence therefore points to a consistent improvement at fixed learning rates rather than a distinct limiting solution, which is what keeps the anchor-protocol comparison central. The improvement is not due to early stopping: extending training to 1.2×10^5 steps shows stable recovery values from a training loss of 10^{-4} down to the stringent 10^{-7} interpolation threshold (mean $0.1694 \rightarrow 0.1691$, with individual seeds shifting by no more than 3×10^{-4} ; Appendix D.8). This trajectory—from coordinate-wise clipping to an unexplained performance ceiling, then to a global-norm correction—mirrors, on a smaller scale, the broader validation of the paper’s symmetry criterion.

10.1 WHERE THE BIAS SUCCEEDS AND WHERE IT DOES NOT

Implicit bias lacks a separate, adjustable strength parameter. Across the three dimensions explored in the FlowAdam setup of §10 (varying observation density, signal conditioning, and explicit regularization, as per Appendix C), a certain pattern emerges regarding the advantage of flow-based or equivariant bias over a well-chosen explicit weight decay: the former demonstrates clear benefits only when the optimal amount of regularization is relatively low and structured. When the signal is dense and well-conditioned, the method consistently outperforms its Adam equivalent at the same L2 placement, achieving gains of around 3–6% for all five random seeds in our experiments (results not tabulated). On the other hand, in the more difficult regime, where strong regularization is necessary (e.g. with sparse observations or ill-conditioned, low-rank targets), using the flow hurts performance compared to training with Adam directly.

Three more limitations stem from the theoretical properties of the described procedure rather than from empirical observation. First, phenomena such as grokking (Power et al., 2022) likely require persistent forces that have no analogue in the geometric flow, such as weight decay, to bias the trajectory after task gradients have vanished. Second, in physics-informed neural networks (PINNs) (Raissi et al., 2019), stiff problems have gradients whose magnitudes differ vastly across coordinates (Wang et al., 2021). There, anisotropic (Adam-style) optimization is more beneficial than isotropic (SGD-style) optimization, as recently advocated in ℓ_∞ geometry (Xie & Li, 2024; Gronich & Vardi, 2026). Third, LoRA fine-tuning (Hu et al., 2022) factors the adapter weight matrix into lower-rank matrices, i.e. it introduces a rank constraint. By fixing the rank rather than learning it, LoRA breaks the mechanism discussed in this paper; it may, however, still find better solutions inside the lower-dimensional subspace if they exist.

11 DISCUSSION AND LIMITATIONS

Scale. The original experiments were deliberately modest in scope, using deterministic twins, matrices of modest size (40×40 to 2000×200) and transformers of up to six layers, since establishing the mechanism required focusing on determinism and interpolation discipline rather than on raw size. To ground these claims, we scaled up a replication experiment on H100 GPUs, varying the problem size from $n=64$ to $n=256$ and using ten different random seeds (Appendix D.9). We tested attention twins with four and six layers and one character language model (Appendix D.5), and applied the method to a second hyperspectral dataset (§9). On the sensing ladder, gradient descent, scalar-Adam, and Muon significantly outperformed the coordinate-wise Adam-family methods at every size, with Muon exact up to $n=256$. On the hyperspectral scenes only the GD–Adam ordering survives: Muon loses to Adam on Indian Pines at both densities, and scalar-Adam’s grid there is undersampled (§9, Appendix D.6). The attention gauge-split was generally scalable and robust to realistic stochastic training, and similar dynamics were observed in the second dataset.

Two exceptions we left untuned. The distinctive two-cluster structure in Figure 1 is qualified by these two exceptions. Shampoo with damping ($\lambda=1$) is evaluated at scale without parameter tuning, which means its absolute recovery degrades with increasing n : it sits far below Adam at $n=40$, overtakes Adam at $n=64$, and remains above it thereafter, albeit not monotonically—it dips slightly below Adam again at $n=96$. The reason for this is that we kept all hyperparameters at their originally published values for all methods and problem sizes, varying only the learning rate. This means that Adam’s $(\beta_1, \beta_2, \epsilon)$, RMSProp’s decay hyperparameter, and Muon’s Newton–Schulz coefficients were kept as published. Tuning λ in Shampoo only at the sizes where it misbehaves would have amounted to outcome-based tuning, which §3 advises against; hence we note its deterioration across scales instead. A similar consideration applies to the long-anneal variant of signum in Appendix D.10, which drives a coordinate-wise method to zero recovery. In both cases the relative performance ordering of GD, scalar-Adam, and Muon remains the same, but taken together these two exceptions make the zoo’s empty gap a statement about this computational budget and this size range rather than about an asymptotic principle. Two further points: the $n=384$ case probably has a better learning rate to be found, as Muon did not reach its interpolating rate on the grid we ran there; and several methods’ recovery values are budget-restricted estimates, with a decent amount of room for improvement from a wider hyperparameter search.

Not a prescription for practice. With this, we think the results neither single out a particular method to be favored in practice nor suggest that GD is somehow better than Adam. In fair comparisons, with the same budget spent on tuning, no optimizer has been found to consistently outperform a well-tuned Adam (Schmidt et al., 2021; Dahl et al., 2023). It is important to note, finally, that what we compare is not how quickly a method optimizes the training objective, but which interpolant it selects at a fixed budget. Adam has also been shown to be more robust to issues such as gradient heterogeneity and heavy-tailed class imbalance (Kunstner et al., 2024), neither of which appears in this task.

Open theoretical directions. Equivariance has been the geometric condition that allowed us to analyze the GD family, but the transfer theorem only applies to the common-scalar subclass (Remark 4.7). While equivariance alone does not specify the interpolant—ScaledGD is equivariant but equalizes the schedule (Remark 4.7)—the non-equivariant annealed sign descent of Appendix D.10 shows that it is not strictly necessary either. Classifying interpolants thus requires consideration of the gauge and of the spectral schedule, and two directions seem natural for extending the theory as it stands. First, a full-dynamics theorem for the boundary in Table 4 (and Figure 4), along the lines of Kang et al. (2026), would characterize the equivariant greedy-schedule regime in which the gradient-flow bias is maintained, with ScaledGD (schedule-equalizing) and signum (flattening, large-step) as its extremes along the two axes. Proposition 8.1 captures this in a decoupled two-factor model, leaving aside the interesting possibility of interacting schedules.

Second, a momentum-invariance lemma. Since the first-moment EMA is linear and respects the gauge, showing that it preserves the limit point of the flow would generalize the observed small-step trend for scalar-Adam into an exact equivalence with gradient flow, and would extend the transfer theorem to the momentum-augmented methods it currently sets aside.

Stochasticity. Our twin models are both full-batch and deterministic. Mini-batch noise adds a diffusion term whose interaction with the gauge (and with Muon’s msign-induced chaos) we have probed only empirically—in the character-language-model twins of §6 and the noise-twin ladder of Appendix D.5—but not characterized. The quotient-geometric framework of Aladrah et al. (2026) is a natural language for that interaction.

Further consequences. The consequence for alignment-based model merging is in §6; Appendix F takes up what the criterion implies for invariant-optimizer design.

Reproducibility. The experiments seeking to understand the mechanism ran on CPU in minutes. Only the H100 replication ladder (Appendix D.9) and the bigger attention twins needed a GPU. The code, random seeds, and all control conditions (including schedule symmetrization, model selection criteria, learning-rate curves, initialization variations, and noise-twin scaling) are available at <https://github.com/idevender/loss-basis-adam>.

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USE OF LARGE LANGUAGE MODELS

LLMs were used for copy-editing only. All questions, proofs, experiments, and numbers are the author’s own and were checked against the derivations and the experiment logs.

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A INSIDE THE EQUIVARIANT CLASS

Theorem 4.5 pins down the shape of an equivariant memoryless rule. This appendix collects what lives inside that shape: the spectral coordinate, the extension to stateful rules and to deeper factorizations, and what happens to balancedness.

Proposition A.1 (The spectral transfer function: the within-class coordinate). *Suppose $H = \psi(GG^\top)$, where ψ is a scalar-valued spectral function applied via functional calculus to the symmetric positive-semidefinite Gram matrix $GG^\top$ —that is, ψ acts as multiplication by $\psi(\lambda)$ on each corresponding eigenspace. Given the singular value decomposition $G = \sum_i \sigma_i u_i v_i^\top$, it follows that*

$$\Phi(G) = \sum_i h(\sigma_i) u_i v_i^\top, \quad \text{where } h(\sigma) := \psi(\sigma^2) \sigma.$$

This means the update progresses along the singular directions of the gradient itself, scaling each by a scalar factor $h(\sigma_i)$, which we refer to as the spectral transfer function. (When an inverse or root of a singular positive-semidefinite matrix appears later, it is interpreted as acting on the support, with zero assigned on the nullspace.) For memoryless one-step methods, standard gradient descent (GD) corresponds to $h(\sigma) = \eta \sigma$, while common-scalar-preconditioned GD gives $h_t(\sigma) = \eta \sigma / a_t$, equivalent to a time-rescaled GD (Theorem 4.6). When applied to a single input matrix, both the exact polar map and support-restricted undamped one-step Shampoo fully flatten nonzero singular values, resulting in $h \equiv \eta$. In contrast, accumulated damped Shampoo, momentum-based methods, and Muon depend on internal state; thus, the above expressions serve as useful one-step approximations rather than full descriptions of their dynamics.

The fractional family introduced by Dong & Sawin (2026), which replaces each σ_i with σ_i^q , directly implements $h(\sigma) = \eta \sigma^q$. Their exponent q is not our dial p , and it runs the other way: $q=1$ is GD and $q=0$ is the polar map, whereas $p=1$ is Adam and $p=0$ is the shared scalar.

Two implications of Proposition A.1 are particularly relevant below.

(i) The ratio $h(\sigma)/h(\sigma_{\max})$, measuring relative amplification of small versus large singular values, defines the “aggressiveness” axis explored in §8. However, this quantity alone does not fully rank practical optimizers: for instance, the one-step idealizations of the exact polar map and undamped Shampoo both yield $h \equiv \eta$, yet in actual trajectories, accumulation and damping cause Shampoo to flatten singular values less aggressively than Muon—an ordering (GD < Shampoo < Muon) based on empirical observation rather than the idealized form.

(ii) Near rank-deficient matrices, the Lipschitz constant of Φ between nearby matrices is at least $|h(\delta)|/\delta$, as seen when comparing diagonal matrices $\text{diag}(1, \pm\delta)$ for small δ (distinct from Adam’s stabilizer ε). For the exact msign operation, this quotient diverges as $\delta \rightarrow 0$ whenever $h(0^+) > 0$, which includes the exact polar map and support-restricted one-step Shampoo. In contrast, GD and scalar-preconditioned GD maintain bounded ratios (η and η/a_t , respectively), provided a_t remains bounded away from zero; rules where the scaling vanishes with the gradient norm (e.g. $a = \|G\|_F$) are excluded.

The practical finite Newton–Schulz iteration is equivariant under real arithmetic and Lipschitz continuous. Using the standard normalization $X_0 = M/(\|M\|_F + \varepsilon_{\text{NS}})$ (with small stabilizer ε_{NS} , NS denoting Newton–Schulz) and first coefficient $a = 3.4445$ as in Jordan et al., 2024 (see Appendix C), after five iterations the slope at zero singular values in original units becomes $a^5/(\|M\|_F + \varepsilon_{\text{NS}})$, conditioned on the global scale rather than a fixed univariate h . As a result, the finite map can greatly enhance small singular values while not introducing the discontinuity of the exact counterpart. This behavior is fundamentally different from the effect of breaking the gauge symmetry, and the two can be distinguished experimentally using the twin protocol of §6. Remark A.2 generalizes this observation to state-dependent update rules.

Remark A.2 (Stateful rules and right preconditioners). With state-dependent update rules, the transformation properties of the various objects are still consistent: momenta transform covariantly ($M_t \mapsto M_t Q$), Frobenius norms and left-side accumulators $L_t = \sum_u G_u G_u^\top$ remain invariant, while right-side quantities—such as $R_t = \sum_u G_u^\top G_u$, or factor Gram matrices $U^\top U$ and $V^\top V$ —transform via conjugation as $R_t \mapsto Q^\top R_t Q$. A sufficient construction preserving these properties is

$$\Delta = H(\mathcal{J}_t) Z_t K(R_t),$$

where Z_t is a covariant quantity, like G_t or M_t , $\mathcal{I}_t$ is gauge-invariant, and K is a spectral function. This captures the covariant structures present in momentum-based gradient descent ($Z_t = M_t$), scalar-Adam, Muon ($Z_t = M_t$), Shampoo, and ScaledGD. It does not capture all stateful update rules respecting equivariance, however, only those composed as a product of per-factor update rules and their gauge-invariant couplings.

Corollary A.3 (Deeper factorizations). *Assume now a composite weight matrix $W = U_1 U_2 \cdots U_D$ of depth D (written D , not L , which is the objective), so that the loss is invariant under the transformations at each factorization interface,*

$$(U_i, U_{i+1}) \mapsto (U_i Q_i, Q_i^\top U_{i+1}), \quad Q_i \in \mathcal{O}(k_i),$$

with Q_i orthogonal. Then gradients transform covariantly at each interface, with a change of basis on the side each factor acts. As a consequence, the earlier covariance and non-covariance conditions on the update rule carry over to each interface individually, with the caveat of transposition for the gradient terms. Any fixed nonlinear coordinate-wise update rule will fail to be equivariant at any interface where the internal dimension satisfies $k_i \geq 2$. On the other hand, if all factor gradients are scaled by the same positive scalar $a(t)$, then the dynamics for all D factors reduce to a single time-rescaled gradient flow, and the transfer theorem continues to hold under its original conditions. Equivariance at individual interfaces therefore does not by itself imply the transfer. The gauge freedom of the type discussed in §6 is obtained when $D = 2$.

Proposition A.4 (Balancedness: a special property of GD-type flows, not a general invariance). *For standard gradient flow, and in particular in the setting of Theorem 4.6, the quantity*

$$B_t = U_t^\top U_t - V_t^\top V_t$$

is conserved during training. For diagonally preconditioned flows of the form $\dot{U} = -D_U \odot \nabla_U L$ (and analogously for V), it is conserved when D_U and D_V are one and the same positive scalar, and drifts otherwise at a rate set by how far each departs from that common scalar. Agreeing up to a factor is not enough: $D_U \equiv 1$, $D_V \equiv 2$ already breaks the cancellation. Nevertheless, conservation of B_t is not necessary for good low-rank recovery: neither Muon nor Shampoo conserves B_t , and both carry large final imbalances $\|B_T\|_F$ (1.47–1.65) next to gradient descent’s 0.06 (Table 2), yet both still recover well on the zoo task. The imbalance present at the 10^{-3} initialization is negligible at this precision, so those final values are accumulated drift. In particular, this suggests that balancedness is a property of flows with common-scalar preconditioning, not of equivariant optimization procedures in general.

Remark A.5 (Comments on discretization and step-size choice). Theorem 4.6 concerns continuous-time dynamics, and while the corresponding discrete-time statements hold under smoothness and stability assumptions, an explicit Euler discretization of the memoryless common-scalar flow is only guaranteed to track the trajectory up to an $\mathcal{O}(\eta)$ error over a fixed time interval. In practice, this does not mean that decreasing the step size η always improves performance, nor does it apply directly to scalar-Adam-style algorithms with fixed β . The learning-rate trends of Appendix D.2 are specific to this task and empirical: both GD and scalar-Adam recover better at the smaller rates tested ($0.131 \rightarrow 0.113$ for GD and $0.201 \rightarrow 0.165$ for scalar-Adam), while Adam’s recovery is almost insensitive to the choice of η , staying within 0.56–0.58 across its grid.

B PROOFS

B.1 LEMMA 3.2 (MAXIMAL ISOMETRIC GAUGE)

Proof. For $A \in \text{GL}(k)$,

$$(UA)(VA^{-\top})^\top = UAA^{-1}V^\top = UV^\top,$$

so the full action preserves the represented matrix. If it is an isometry for every (U, V) , take $V = 0$ and let U have an arbitrary single nonzero row $u \in \mathbb{R}^{1 \times k}$. Then $\|uA\|_2 = \|u\|_2$ for every u , hence $AA^\top = I$. Since A is square, this is equivalent to $A \in \mathcal{O}(k)$. Conversely, if $A \in \mathcal{O}(k)$, both right multiplications preserve Frobenius norms, so the action is an isometry. $\square$

B.2 LEMMA 4.1 (GRADIENT COVARIANCE)

Proof. Write $L(U, V) = f(UV^\top)$ and let $G = \nabla f(W) \in \mathbb{R}^{n \times n}$ at $W = UV^\top$. Then $\nabla_U L = GV$ and $\nabla_V L = G^\top U$. At the transformed point (UQ, VQ) the represented matrix is unchanged, $W' = UQ(VQ)^\top = W$, so $\nabla f(W') = G$ and $\nabla_U L(UQ, VQ) = G(VQ) = (GV)Q = (\nabla_U L)Q$; likewise $\nabla_V L(UQ, VQ) = G^\top(UQ) = (\nabla_V L)Q$. $\square$

B.3 PROPOSITION 4.2 (EQUIVARIANT CLASS)

Proof. Throughout, primed quantities denote the run initialized at (U_0Q, V_0Q) , with the state initialized by σ_Q as constructed below. We show by induction that

$$U'_t = U_tQ, \quad V'_t = V_tQ,$$

with the state transforming as stated. By Lemma 4.1, if the iterates are gauge-related at time t , then the factor gradients satisfy $G'_t = G_tQ$.

(1) *GD and momentum.* For GD,

$$U'_{t+1} = U'_t - \eta G'_t = (U_t - \eta G_t)Q.$$

Momentum buffers are linear in the gradients, so by induction (with $\sigma_Q : M \mapsto MQ$),

$$M'_t = \beta M'_{t-1} + G'_t = M_tQ,$$

and the update again right-multiplies by Q . For Nesterov, the gradient is evaluated at the lookahead point $U_t + \beta(U_t - U_{t-1})$. In the primed run this is the same point right-multiplied by Q , and Lemma 4.1 applies there verbatim. Hence the lookahead gradient is also covariant, and the induction goes through unchanged.

(2) *Scalar-preconditioned Adam.* The first moments transform as $M'_t = M_tQ$, since the EMA is linear. The second-moment state is the single scalar

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) \text{mean}(G_t^{\odot 2}), \quad v_0 = 0,$$

where the mean is taken over all entries of both factors. Because the mean is linear, it commutes with the EMA, so this scalar equals $\text{mean}(v_t)$ for the entrywise EMA v_t of $g^{\odot 2}$. Storing the scalar therefore changes no number. It is also what makes the state map literal, since the entrywise v_t does not itself transform under a fixed map (squaring and right-rotation do not commute). Now, the entry mean of $G^{\odot 2}$ is $\|G\|_F^2$ divided by the entry count, so v_t depends on the gradients only through their Frobenius norms. Since $\|G_tQ\|_F = \|G_t\|_F$, we get $v'_t = v_t$. The bias-corrected read $\hat{v}_t = v_t/(1 - \beta_2^t)$ divides by a fixed scalar and is therefore invariant as well. The state map is

$$\sigma_Q : (M_U, M_V, v) \mapsto (M_UQ, M_VQ, v),$$

with $\sigma_Q(0) = 0$ as Definition 3.1 requires. Finally, the update $-\eta \hat{M}_t/(\sqrt{\hat{v}_t} + \varepsilon)$ is a scalar multiple of a covariant matrix, hence covariant.

(3) *Muon.* With momentum $M'_t = M_tQ$ as in (1), it suffices to show that $\text{msign}(MQ) = \text{msign}(M)Q$. Let $M = A\Sigma B^\top$ be an SVD. Then $MQ = A\Sigma(Q^\top B)^\top$ is an SVD of MQ , so

$$\text{msign}(MQ) = A(Q^\top B)^\top = AB^\top Q = \text{msign}(M)Q.$$

Next, consider the Newton–Schulz iteration $X_{j+1} = aX_j + b(X_jX_j^\top)X_j + c(X_jX_j^\top)^2X_j$ with $X_0 = M/(\|M\|_F + \varepsilon_{\text{NS}})$. If $X_j \mapsto X_jQ$, then $X_jX_j^\top$ is invariant, so every term right-multiplies by Q , and the property is preserved at every iteration and at any truncation. Equivariance is therefore exact in real arithmetic for the finite practical algorithm, not only for the idealized msign ; finite-precision sensitivity is a separate issue.

(4) *Shampoo.* The left accumulator $L_t = \sum_{u \leq t} G_u G_u^\top$ is invariant, since $G_u Q Q^\top G_u^\top = G_u G_u^\top$, while $R_t = \sum G_u^\top G_u$ transforms as $R'_t = Q^\top R_t Q$. For any spectral function h we have $h(Q^\top R Q) = Q^\top h(R) Q$. Taking $h(R) = (R + \lambda I)^{-1/4}$, with the damping included since $Q^\top R Q + \lambda I = Q^\top (R + \lambda I) Q$, we obtain

$$\begin{aligned} (L'_t + \lambda I)^{-1/4} G'_t (R'_t + \lambda I)^{-1/4} &= (L_t + \lambda I)^{-1/4} G_t Q Q^\top (R_t + \lambda I)^{-1/4} Q \\ &= [(L_t + \lambda I)^{-1/4} G_t (R_t + \lambda I)^{-1/4}] Q. \end{aligned}$$

$\square$

B.4 PROPOSITION 4.3 (COORDINATE-WISE RULES BREAK THE GAUGE)

Proof. Consider a memoryless entrywise map $\Phi(G)_{ij} = \phi(G_{ij})$. Equivariance for all $Q \in O(k)$ requires $\Phi(GQ) = \Phi(G)Q$ for all G . Take $k \geq 2$ and let Q_θ be a rotation by θ in the first two coordinates. Apply the requirement to matrices G supported on a single row $(x, y, 0, \dots)$: it reads

$$\phi(x \cos \theta - y \sin \theta) = \phi(x) \cos \theta - \phi(y) \sin \theta, \quad \phi(x \sin \theta + y \cos \theta) = \phi(x) \sin \theta + \phi(y) \cos \theta$$

for all x, y, θ . Taking $x = y = 0$ in the first identity gives $\phi(0) = \phi(0)(\cos \theta - \sin \theta)$ for every θ , hence $\phi(0) = 0$ (take $\theta = \pi/2$). Setting $y = 0$ in both identities then gives $\phi(x \cos \theta) = \phi(x) \cos \theta$ and $\phi(x \sin \theta) = \phi(x) \sin \theta$; as θ ranges over $[0, 2\pi)$ its cosine and sine sweep all of $[-1, 1]$, so $\phi(cx) = c\phi(x)$ for every $c \in [-1, 1]$ and every $x \in \mathbb{R}$. For any x and any u with $0 < |u| \leq |x|$, choosing $c = u/x \in [-1, 1]$ gives $\phi(u) = (u/x)\phi(x)$, i.e. $\phi(u)/u = \phi(x)/x$. Since any two nonzero reals are both dominated in absolute value by a common third real, the ratio $\phi(x)/x$ takes a single value c on $\mathbb{R} \setminus \{0\}$, whence $\phi(x) = cx$ for all x : linearity, with no appeal to additivity. Thus any nonlinear fixed entrywise map fails equivariance already for a 2×2 rotation. The standard zero-state first steps of Adam and RMSProp are nonlinear maps of this form, as are signSGD and Lion’s outer-sign update. An explicit counterexample with $\theta = \pi/4$ is immediate (with the usual convention $\text{sign}(0) = 0$). Take the single row $G = (1, 1)$, so $GQ_{\pi/4} = (0, \sqrt{2})$. Applying sign after the rotation gives $\text{sign}(GQ_{\pi/4}) = (0, 1)$, whereas rotating the sign image gives $\text{sign}(G)Q_{\pi/4} = (1, 1)Q_{\pi/4} = (0, \sqrt{2})$; the two disagree, so sign is not gauge-equivariant.

Adafactor. Its second-moment surrogate is not entrywise but rank-one factored, so we check it directly. Adafactor replaces the per-coordinate second moment by the reconstruction $\hat{v}_{ij} = r_i c_j / \sum_l r_l$ from the row and column sums $r_i = \sum_j v_{ij}$, $c_j = \sum_i v_{ij}$ of the accumulator $v = \text{EMA}(g^2)$, with update $g_{ij} / \sqrt{\hat{v}_{ij}}$. Taking a single step (so that $\hat{v} = g^{\odot 2}$ after bias correction; any common positive factor here rescales both sides of the test alike), the row sums are gauge-invariant, $r_i = (GG^\top)_{ii}$, but the column sums are the basis-dependent diagonal $c_j = (Q^\top G^\top G Q)_{jj}$ of the conjugated Gram: already the $\theta = \pi/4$ rotation of $G = \text{diag}(1, \delta)$ with $0 < \delta \neq 1$ leaves $r = (1, \delta^2)$ fixed while sending $c = (1, \delta^2) \mapsto \frac{1}{2}(1 + \delta^2)(1, 1)$. Writing $Q = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, the resulting zero-state updates are

$$\Phi(G) = \sqrt{1 + \delta^2} \text{diag}(1, \delta^{-1}), \quad \Phi(GQ) = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix},$$

whereas

$$\Phi(G)Q = \frac{\sqrt{1 + \delta^2}}{\sqrt{2}} \begin{pmatrix} 1 & \delta^2 \\ -\delta^{-1} & \delta^{-1} \end{pmatrix} \neq \Phi(GQ).$$

The two sides differ at order 1 for fixed δ , while a small scalar stabilizer added to $\hat{v}$ perturbs both sides continuously, so a sufficiently small stabilizer does not reconcile them. Adafactor therefore breaks equivariance through a factored, coordinate-dependent second moment. $\square$

B.5 PROPOSITION 4.4 (EXACT FIRST-STEP ADAM DEFECT)

Proof. For either factor and its gradient G , zero initial state and bias correction give

$$m_1 = (1 - \beta_1)G, \quad v_1 = (1 - \beta_2)G^{\odot 2}, \quad \hat{m}_1 = G, \quad \hat{v}_1 = G^{\odot 2}.$$

Hence the first Adam update is exactly $-\eta D_\epsilon(G)$; in particular, it is independent of β_1, β_2 under the stated convention. By Lemma 4.1, the gradients in the rotated run are $\tilde{G}_U = G_U Q$ and $\tilde{G}_V = G_V Q$. Gauge-align its first iterate back to the reference coordinates:

$$\tilde{U}_1 Q^\top = U_0 - \eta D_\epsilon(G_U Q) Q^\top = U_0 - \eta [D_U + E_Q(G_U)],$$

and likewise

$$\tilde{V}_1 Q^\top = V_0 - \eta [D_V + E_Q(G_V)].$$

The gauge-aligned and reference products are

$$\tilde{W}_1 = (\tilde{U}_1 Q^\top)(\tilde{V}_1 Q^\top)^\top,$$

$$W_1 = (U_0 - \eta D_U)(V_0 - \eta D_V)^\top.$$

Subtracting and expanding gives the stated identity. The alternative quadratic expression follows from $D_\epsilon(GQ)Q^\top = D_\epsilon(G) + E_Q(G)$ for each factor. For the displayed witness, $G_U = G_V = (1, 0)$ while $G_U Q = G_V Q = (2^{-1/2}, 2^{-1/2})$; direct substitution gives the two scalar products in the statement. $\square$

B.6 THEOREM 4.5 (STRUCTURE)

Proof. $(\Leftarrow) (GQ)(GQ)^\top = GG^\top$, so $\Phi(GQ) = H(GG^\top)GQ = \Phi(G)Q$. $(\Rightarrow)$ For full-column-rank G let $G^+ = (G^\top G)^{-1}G^\top$ (the Moore–Penrose pseudoinverse for full column rank) and define $X(G) := \Phi(G)G^+ \in \mathbb{R}^{n \times n}$. Then $X(G)G = \Phi(G)(G^+G) = \Phi(G)$, and X is gauge-invariant: $(GQ)^+ = Q^\top G^+$, so $X(GQ) = \Phi(G)QQ^\top G^+ = X(G)$. It remains to see that X depends on G only through $GG^\top$, i.e., that full-column-rank matrices with equal Grams lie on one gauge orbit: if $G_1G_1^\top = G_2G_2^\top$, set $Q := G_1^+G_2$; then $Q^\top Q = G_2^\top(G_1G_1^\top)^+G_2 = G_2^\top(G_2G_2^\top)^+G_2 = I_k$ (full column rank), and $G_1Q = G_1G_1^+G_2 = P_{\text{range}(G_1)}G_2 = G_2$ since $\text{range}(G_1)$ and $\text{range}(G_1G_1^\top)$ coincide. Hence $X(G_1) = X(G_2)$, and $H(GG^\top) := X(G)$ is well defined, giving $\Phi(G) = H(GG^\top)G$. In the square invertible case, the polar decomposition $G = (GG^\top)^{1/2}O$ with $O \in O(n)$ gives directly $\Phi(G) = \Phi((GG^\top)^{1/2})O = H(GG^\top)G$ with $H(P) = \Phi(P^{1/2})P^{-1/2}$; uniqueness follows by evaluating at $G = P^{1/2}$. $\square$

B.7 PROPOSITION A.1 (SPECTRAL TRANSFER)

Proof. With the compact SVD $G = \sum_i \sigma_i u_i v_i^\top$ we have $GG^\top = \sum_i \sigma_i^2 u_i u_i^\top$, so $\psi(GG^\top) = \sum_i \psi(\sigma_i^2) u_i u_i^\top + \psi(0)P_0$, where $P_0 = I - \sum_i u_i u_i^\top$ projects onto $\ker(GG^\top)$ (nontrivial whenever G is rank-deficient, in particular when $k < n$). Since $P_0G = 0$ that term drops out, and $\psi(GG^\top)G = \sum_i \psi(\sigma_i^2) \sigma_i u_i v_i^\top$. The instances follow by inspection; for support-restricted one-step undamped Shampoo,

$$(GG^\top)_{\text{supp}}^{-1/4} G (G^\top G)^{-1/4} = \sum_i \sigma_i^{-1/2} \sigma_i \sigma_i^{-1/2} u_i v_i^\top = \sum_i u_i v_i^\top = \text{msign}(G).$$

For the Lipschitz claim (consequence (ii) stated after the proposition), fix $0 < \delta < 1$ (a small singular value, distinct from Adam’s stabilizer ε) and evaluate at $G_\pm = \text{diag}(1, \pm\delta)$ (embedded in the top-left 2×2 block): the second singular pair of $G_\pm$ is $(\sigma, u, v) = (\delta, e_2, \pm e_2)$, so $\Phi(G_+) - \Phi(G_-) = 2h(\delta)e_2e_2^\top$ while $G_+ - G_- = 2\delta e_2e_2^\top$; the corresponding Lipschitz quotient is at least $|h(\delta)|/\delta$, which diverges as $\delta \rightarrow 0$ whenever $h(0^+) > 0$, and equals η for $h(\sigma) = \eta\sigma$. $\square$

B.8 COROLLARY A.3 (DEEPER FACTORIZATIONS)

Proof. Write $L(U_1, \dots, U_D) = f(U_1 \cdots U_D)$, fix an interface i , and set $A = U_1 \cdots U_{i-1}$, $B = U_{i+2} \cdots U_D$, $G = \nabla f(W)$. Then $\nabla_{U_i} L = A^\top G (U_{i+1} B)^\top$ and $\nabla_{U_{i+1}} L = (A U_i)^\top G B^\top$. Under $(U_i Q, Q^\top U_{i+1})$ the represented W , hence G , is unchanged, so $\nabla_{U_i} L \mapsto (\nabla_{U_i} L) Q$ and $\nabla_{U_{i+1}} L \mapsto Q^\top (\nabla_{U_{i+1}} L)$: the gradient transforms covariantly on the acted side of each factor, which is the only property the cited proofs use (transposed for the left action). $\square$

B.9 THEOREM 4.6 (TRANSFER THEOREM)

Proof. Let $\theta(t)$ be a locally absolutely continuous solution of $\dot{\theta} = -\nabla L(\theta)/a(t)$ with $a(t) > 0$ determined by gauge-invariant trajectory statistics. Define $\tau(t) = \int_0^t du/a(u)$, a strictly increasing bijection onto $[0, \tau_{\max})$ with $\tau_{\max} = \int_0^\infty du/a(u)$, and $\tilde{\theta}(\tau) = \theta(t(\tau))$. Then, for a.e. τ , $\frac{d\tilde{\theta}}{d\tau} = \dot{\theta} \cdot \frac{dt}{d\tau} = (-\nabla L(\theta)/a) \cdot a = -\nabla L(\tilde{\theta})$: $\tilde{\theta}$ is a Carathéodory gradient-flow solution with the same initial condition, and the two trajectories traverse the same set of points under a monotone reparameterization: the whole gradient-flow path when $\tau_{\max} = \infty$, and otherwise its prefix $\tilde{\theta}([0, \tau_{\max}))$. So any property of the gradient-flow path on the traversed portion holds for the scalar-preconditioned flow. When $\tau_{\max} = \infty$, which holds in particular whenever a is bounded above (since then $\int^\infty du/a(u)$ diverges), the reparameterization also reaches the gradient-flow limit, so the limit-point characterizations of Gunasekar et al. (2017); Arora et al. (2019); Li et al. (2021) under their hypotheses transfer as well; if $\tau_{\max} < \infty$ only the finite-time path is shared. Since a is computed from gauge-invariant statistics, the reparameterization is itself gauge-independent, so the statement holds uniformly over the gauge orbit. $\square$

B.10 PROPOSITION A.4 (BALANCEDNESS)

Proof. Under gradient flow, $\dot{U} = -GV$, $\dot{V} = -G^\top U$ with $G = \nabla f(W)$:

$$\frac{d}{dt}(U^\top U) = -V^\top G^\top U - U^\top GV, \quad \frac{d}{dt}(V^\top V) = -U^\top GV - V^\top G^\top U,$$

which are equal, so $\dot{B} = 0$. (Time reparameterization by $1/a(t)$ multiplies both by the same scalar; conservation persists for the whole class of Theorem 4.6.) Under a diagonal preconditioned flow $\dot{U} = -D_U \odot (GV)$, $\dot{V} = -D_V \odot (G^\top U)$,

$$\dot{B} = -[(D_U \odot GV)^\top U + U^\top (D_U \odot GV)] + [(D_V \odot G^\top U)^\top V + V^\top (D_V \odot G^\top U)].$$

Write $D_U = c\mathbf{1}\mathbf{1}^\top + E_U$ and $D_V = c\mathbf{1}\mathbf{1}^\top + E_V$ for a *common* scalar c , where $\mathbf{1}\mathbf{1}^\top$ is the all-ones matrix of the factor's shape. The common-scalar terms cancel, while the remainder obeys

$$\|\dot{B}\|_F \leq 2\|E_U\|_\infty \|GV\|_F \|U\|_F + 2\|E_V\|_\infty \|G^\top U\|_F \|V\|_F,$$

by $\|E \odot Z\|_F \leq \|E\|_\infty \|Z\|_F$ (with $\|\cdot\|_\infty$ the entrywise maximum) and the Frobenius submultiplicativity bound. Thus, at bounded factor and gradient norms, the drift vanishes as both factor preconditioners approach the same scalar. $\square$

B.11 PROPOSITION 8.1 (GREEDY VS. EQUAL-RATE PHASE BOUNDARY)

Proof. Each decoupled mode is a balanced two-factor scalar factorization $w = uv$ with $u = v = \sqrt{w}$ and per-mode loss $\frac{1}{2}(uv - s)^2$, so $\partial_u L = v(w - s)$ and $\partial_v L = u(w - s)$. The balanced set $u = v$ is invariant under both flows below, since $\dot{u} = \dot{v}$ there, so the reduction to the single variable w is consistent. (The tied model $w = u^2$ gives identical dynamics at twice the rate in (i), a uniform time-rescaling.)

(i) The $h(\sigma) = \eta\sigma$ (gradient) flow is $\dot{u} = -\eta v(w - s)$, $\dot{v} = -\eta u(w - s)$, hence $\dot{w} = \dot{u}v + u\dot{v} = -\eta(u^2 + v^2)(w - s) = 2\eta w(s - w)$ (using $u^2 = v^2 = w$ at balance): logistic, with solution $w(t) = s(1 + (s/w_0 - 1)e^{-2\eta st})^{-1}$. The head-fit condition $w_1(T_1) = (1 - \delta)s_1$ gives $e^{2\eta s_1 T_1} = \frac{1 - \delta}{\delta}(s_1 - w_0)/w_0 \leq \frac{1 - \delta}{\delta}s_1/w_0$. For the tail, the exact solution obeys $w(t) \leq \frac{s w_0}{s - w_0} e^{2\eta st} \leq 2w_0 e^{2\eta st}$ for $w_0 \leq s/2$, so

$$w_2(T_1) \leq 2w_0(e^{2\eta s_1 T_1})^\rho \leq 2\left(\frac{1 - \delta}{\delta}\right)^\rho s_1^\rho w_0^{1 - \rho} = C_{\delta, \rho} s_1 \left(\frac{w_0}{s_1}\right)^{1 - \rho}.$$

(ii) The $h \equiv \eta$ flow acts on every factor by replacing its update with η times the matrix sign of its gradient. In the scalar case this gives $\dot{u} = -\eta \text{sign}(v(w - s)) = \eta$, so that u grows linearly in time, and likewise $\dot{v} = \eta$, for $0 < w < s$. Hence each of the two factors increases like $\sqrt{w_0} + \eta t$, i.e. $\sqrt{w} = \sqrt{w_0} + \eta t$ until w reaches the threshold s ; there the gradient becomes zero, and since $\text{sign}(0) = 0$ (as also follows from $\text{msign}(0) = 0$ by Proposition 4.2(3)), the mode stops. So the capped dynamics are the stopped flow. Mode i therefore fits at $t_i = (\sqrt{s_i} - \sqrt{w_0})/\eta$, and the head reaches $(1 - \delta)s_1$ at $T_1 = (\sqrt{(1 - \delta)s_1} - \sqrt{w_0})/\eta$. For $\delta < 1 - s_2/s_1$ we have $(1 - \delta)s_1 > s_2$, hence $t_2 < T_1$ and $w_2(T_1) = s_2$ exactly, with no dependence on w_0 . $\square$

C EXPERIMENTAL DETAILS

Sensing task (§5–§7, §8).

- *Target and measurements.* $X^* = U^*V^{*\top}$, $U^*, V^* \in \mathbb{R}^{40 \times 3}$ i.i.d. Gaussian scaled by $1/\sqrt{3}$; $m = 462 = 2 \text{ dof}$, $\text{dof} = 3(80 - 3) = 231$; A_i i.i.d. standard Gaussian; loss = $\text{mean}_i(\langle A_i, W \rangle - y_i)^2$.
- *Factors.* $U, V \in \mathbb{R}^{40 \times 40}$ initialized $\mathcal{N}(0, 10^{-6})$ entrywise (10^{-3} scale).
- *Budget.* 2×10^4 steps (extended where noted); interpolation bar 10^{-7} (checked every 200 steps); 3 paired seeds {42, 123, 456}.

Optimizer hyperparameters:

- *Adam, scalar-Adam.* $\beta = (0.9, 0.999)$, $\varepsilon = 10^{-8}$.

- *RMSProp*. Classical form, second-moment decay 0.99 and no first moment, $\varepsilon = 10^{-8}$.
- *Muon*. Momentum 0.9, Newton–Schulz 5 iterations, coefficients (3.4445, -4.7750 , 2.0315) (Jordan et al., 2024).
- *Shampoo*. Accumulated (non-EMA) L, R , damping $\lambda = 1$, inverse-root refresh every 20 steps.
- *Adafactor*. Row/column EMA factored second moment.
- *Lion*. $\beta_1 = 0.9$ inner, 0.99 EMA.
- *signum*. Sign of an EMA momentum with $\beta_1 = 0.9$ (Adam’s β_1) and no second moment, no clipping.

Learning-rate grids per method span $\geq$ one decade around the interpolating range (curves for five reference methods in Appendix D). The dial’s fixed-step arm (§7) re-runs the full p -sweep at each single rate of the shared grid $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$, same update rule, seeds, and interpolation bar (experiments/precond_dial_fixed_lr.py).

Both dial arms use a 3×10^4 -step budget (experiments/precond_dial_scalar_check.py), longer than the zoo’s 2×10^4 . That CPU implementation follows Algorithm 1 line by line; two other code paths place ε differently. The GPU port behind the $n=128$ dial rows (experiments/nibi/common.py) omits the outer ε for $0 < p < 1$ and divides by $\bar{s} + \varepsilon$ rather than $\bar{s} + 2\varepsilon$ at $p=0$. The released flowadam package, used for the FlowAdam- p rows, folds the bias correction into the step size at $p=1$ and adds ε to $\sqrt{v}$ before that factor—TensorFlow’s Adam convention rather than Algorithm 1’s—for an effective $\varepsilon/\sqrt{1-\beta_2^t}$. At $\varepsilon = 10^{-8}$ neither departure reaches the digits we report. Its shared scalar for $p < 1$ is the geometric mean rather than Algorithm 1’s RMS, so the FlowAdam- p rows sit on the geometric-mean dial of §7’s footnote; §10 reads their gain against the RMS dial’s 0.201 rather than the geometric-mean endpoint 0.229, which is the conservative of the two.

Attention task (§6).

- *Task*. Modular addition modulo 47 on input sequences of the form $(a, b, =)$, with 40% of the data used for training (fixed split).
- *Model*. A 2-layer transformer with 4 attention heads, model dimension $d_{\text{model}} = 64$, using RMSNorm (Zhang & Sennrich, 2019) in pre-normalization configuration and SiLU-activated (Elfwing et al., 2018) MLPs of width 256. The model omits biases and does not apply query–key normalization (qk-norm), so the gauge holds exactly.
- *Training*. Full-batch optimization over 1500 steps, executed deterministically on CPU (torch.use_deterministic_algorithms).
- *Gauge*. An orthogonal transformation A_h per head, derived via QR decomposition of a Gaussian matrix, applied to the row-blocks of W_Q and W_K .
- *Learning rates*. Adam uses 10^{-3} ; SGD with heavy-ball momentum (0.9, unnormalized, as for Muon’s momentum here) uses 0.5; scalar-Adam is set to 3×10^{-3} , while Muon uses 0.02 on matrix parameters, with the standard hybrid applying Adam at 10^{-3} to the embeddings and the output head.

Drift is defined as the relative ℓ_2 logit distance on the validation set, $\|Z - Z'\|_F / \frac{1}{2}(\|Z\|_F + \|Z'\|_F)$, where Z and Z' are the validation logit matrices of the two model copies. QK drift is computed the same way, but on $W_Q^\top W_K$ for each attention head and then aggregated across heads. The symmetrized ratio is used in both cases.

Hyperspectral task (§9).

- *Scenes*. The Indian Pines dataset is the corrected AVIRIS cube (Baumgardner et al., 2015), with 2000 randomly sampled pixels over 200 spectral bands (water-absorption bands removed), and the Pavia University dataset is 2000 pixels in 103 bands from the ROSIS sensor. Both scenes are taken from Graña et al. (2021) with band-wise mean removed.
- *Model and loss*. The factorization rank is 48, at initialization scale 10^{-2} ; no weight decay ($\text{wd} = 0$); entrywise completion loss minimized over the entries in the observed mask.
- *Densities*. Two densities per scene were selected, for which $m/\text{dof}_{24} \approx 1.15$ and 1.9 (corresponding to 0.15/0.25 for Indian Pines and 0.28/0.46 for Pavia).

- *Protocol.* Checkpoints are selected by matching training-set loss values in the range 3×10^{-3} to 10^{-5} ; learning rates are chosen by the train-only selection rule of Appendix D.6. The experiments were repeated with four random seeds (two for the learning-rate selection and two independent runs), with results aggregated across all four. Since the selection procedure only uses the training set, it carries no information about the test set.

The degrees of freedom are counted at rank 24, which is where each scene’s singular spectrum begins to stabilize (leaving a residual energy of about 0.10 for Indian Pines). Hence the ratio m/dof_{24} characterizes how underdetermined the underlying signal is. The model is over-parameterized to rank 48, so that the implicit bias, rather than a rank constraint, selects the solution. Every reading in §9 and Table 5—both scenes, both densities—comes from the same GPU float64 harness under the same train-only rule; the 3×10^{-5} and $\leq 10^{-5}$ figures are that harness read at two depths of fit.

FlowAdam experiments (§10). Following Singh & Sheel (2026), we test two clipping modes, `percoord` (each entry clipped to $[-1, 1]$) and `globalnorm` (the whole velocity rescaled to norm $\leq c$, so c is the threshold referred to in §10), and a varying `precond_power`, on a step budget of 3×10^4 (`experiments/flowadam_upgrade.py`), longer than the zoo’s 2×10^4 ; for reference, the early-stopping audit of Appendix D.8 goes up to 1.2×10^5 . We use the same three paired seeds and the same 10^{-7} interpolation threshold. Two configurations failed to meet that threshold within the allotted steps: FlowAdam- $p=0$ got to 7×10^{-6} , and global-norm clipping at $p=1$ reached 6×10^{-7} . Appendix D.8 continues the former to a true 10^{-7} and finds the same 0.1691; the latter’s 0.220 is a deep fit rather than an interpolating reading, and is treated as such in comparisons. The evaluations were carried out in the dense, well-conditioned regime of §10.1, tuned versus tuned, with the same same-placement L2 control (identical L2 term in the loss for the Adam baseline) across five paired seeds.

Comparing values across tables. Some results are repeated with minor variations. The differences come from different learning-rate grids (finer ones in the diagnostic experiments of Appendix D.3), different numbers of seeds (always at least three), and different step limits—but not from a difference in outcome, and overall the finer grids produce slightly lower values. Two observations stand out. First, the zoo’s selection grids stop above the smallest interpolating learning rate, whereas the diagnostic grids of Appendix D.3 go below it, so a flow-limit reading can fall under the corresponding Table 2 row (Remark A.5): GD 0.113 versus 0.131, scalar-Adam 0.165 versus 0.201, and Adam fixed near 0.56–0.58.

Second, because the dial evaluates p on a single shared grid at a coarser resolution, its endpoint values reflect the rates that grid selects rather than those chosen by the zoo or the ladder. For instance, at $p=1$ the value reads 0.570 compared to the zoo’s 0.5734, and at $p=0$ with $n=128$ it reads 0.40 compared to the ladder’s 0.345. Monotonicity in p remains consistent regardless. Discrepancies smaller than 0.01 across tables stem from grid effects, and we do not highlight them individually.

D THE CONTROL BATTERY

All key results reported in the paper were re-evaluated under the following control conditions. None of them showed meaningful changes beyond noise. The only estimate that became more precise with additional random seeds—the phase boundary τ^* —is noted in Appendix D.1. Two controls actually strengthened the conclusions rather than simply verifying them: the can/cannot framing detailed in Appendix D.2, and the observation regarding RMSProp’s conservative behavior.

D.1 SCHEDULE SYMMETRIZATION

Constant-update-norm methods (Muon, signum, Lion) require decay to interpolate; the concern is that decay itself (small late steps) shifts the bias. We ran every method under three schedules (Table 6): constant (early-stopped at interpolation), cosine (early-stopped), and cosine over the *full* 2×10^4 -step horizon with no early stopping, the exact treatment Muon received, including its small late steps.

The phase diagram of §8 was also redone with an added cosine row for each method. Recovery is virtually identical across all τ for GD, Adam, and Shampoo, to within the noise, and Muon is shifted by no more than 0.02. The relative ordering of the methods is the same under both schedules, constant

Table 6: Recovery under three schedules (3 seeds, best-recovery lr per cell) for five reference methods: the equivariant anchors GD, scalar-Adam, and Shampoo, the coordinate-wise anchor Adam, and the constant-norm case Muon. The split does not move; under a single uniform cosine schedule the complete nine-method classification is again 9/9 (selection-rule test, Appendix D.3).

method	constant	cosine	cosine, full horizon
GD	0.1312	0.1312	0.1312
Adam	0.5734	0.5734	0.5764
scalar-Adam	0.2010	0.2010	0.2010
Muon	(0.0084 [†])	0.0000	0.0000
Shampoo	0.2856	0.2856	0.2856

[†]train-loss floor 6×10^{-2} : Muon does not interpolate at constant lr, which is the reason decay is required. Shown for completeness.

and decaying, with Adam having the worst performance across all six τ in both cases. Increasing the number of random seeds from 3 to 10 brought the estimated crossover point from $\tau^* \approx 0.35$ to $\tau^* \approx 0.2$ ($\approx 4\%$ tail energy), but the method ranking and the schedule-invariance of that ranking still hold.

A separate sweep at $n=128$ reproduces the $\text{GD} < \text{Shampoo} < \text{Adam}$ ordering in the low- τ part of the grid. However, we prefer to report this experiment only qualitatively, because on the ladder’s own per-size grid and at $\tau=0$, untuned Shampoo instead sits above Adam at $n=128$ (Table 11). This is not a contradiction to our main result, since the per-size grid and the procedure are different. Within this sweep, the same untuned damping shows again at higher τ : starting from $\tau=0.25$, Shampoo edges above Adam, by less than 0.01 throughout and well within the seed spread of either method, so unlike at $n=40$, Adam is not universally the worst optimizer at $n=128$. This sweep used a different two-rate grid from the one used in the main ladder experiments (Appendix D.9). On that grid Muon does not recover cleanly at $n=128$, its value being 0.54 at $\tau=0$ instead of 0, so its crossing is not added to the table. It should be noted, nevertheless, that the ladder’s broader grid does intersect Muon’s interpolation band at $n=128$, where Muon indeed recovers fully (Table 11), so the difference is in the coverage, not in the result.

D.2 THE LEARNING-RATE AXIS

Choosing a method’s learning rate based on optimal recovery constitutes selection by outcome, while selecting based on fastest interpolation ignores bias considerations—yet leads each method to its most bias-distorted step size (even Muon degrades to 0.91 recovery at its fastest-interpolating rate). Neither criterion on its own supports a conclusion about implicit bias. We therefore evaluate recovery as a function of learning rate (Appendix Table 7) and discuss an achievability asymmetry: across the tested grids, every equivariant method reaches good recovery at some learning rate, whereas no coordinate-wise approach does at any rate, even its best (Adam’s recovery stays confined to 0.56–0.58 over two orders of magnitude).

Three facts from the experiment support this reading. (i) Adam demonstrated almost no dependence of recovery on the learning rate: across all five tested values it varied only from 0.56 to 0.58. These five learning rates span two orders of magnitude and every one of them interpolates, so the range contains no step size at which Adam keeps the low-rank bias. (ii) GD and scalar-Adam perform better at smaller step sizes (recovery improves from 0.131 to 0.113 for GD, and from 0.201 to 0.165 for scalar-Adam, at their smallest interpolating rates). This aligns with expectations from the common-scalar flow model, though the transfer theorem does not formally guarantee the same behavior for EMA-based scalar-Adam. (iii) Muon displays the reverse trend: it recovers exactly at larger decayed rates (0.0000 error at learning rates 0.03–0.1), but degrades at smaller ones (0.75–0.91 recovery). On this task, that behavior depends on the learning schedule, and it is consistent with equal-rate spectral growth (Kang et al., 2026) and the sensitivity of finite Newton–Schulz iterations near small singular values. Thus, in this setting, equivariance defines a stable class membership, while the learning schedule and step size affect recovery performance within that class.

D.3 LEARNING-RATE CURVES AND SELECTION RULES

The grids of Table 7 extend to lower rates than the zoo’s selection procedure used, and the $\rightarrow \text{fl}$ rates were run specifically to probe the flow limit; they therefore usually lie below the rates selected by the

zoo’s best-recovery criterion (§5), which explains the discrepancy with Table 2. Each zoo method appears in the table at the rate chosen for it by that criterion (GD 0.131 at lr 0.01, Adam 0.573 at 0.01, scalar-Adam 0.201 at 0.001; see the grid note in Appendix C). The reading is the one given in Appendix D.2: Adam is flat—no learning rate preserves the bias—while GD and scalar-Adam improve steadily as they approach the flow limit, and Muon is driven by its large decaying steps (a schedule-driven bias). The remaining coordinate-wise methods (RMSProp, Lion, Adafactor, and signum) were swept on their own grids but are omitted from the table, as they all fall into the poor-recovery category (≥ 0.42) under both selection criteria, as the stress test below reports.

Table 7: Recovery versus learning rate for the five reference methods, averaged over three random seeds. For each method the results occupy two rows: the top row gives the learning-rate grid used (of varying length, since Adam has five rates while the rest have four), and the second row the corresponding recovery, so a method’s row simply ends where its grid does. “ $\rightarrow$ fl” signifies the lowest learning rate at which interpolation still succeeds, the flow-limit-aligned column; n/i = does not interpolate; div = diverges.

method		learning-rate grid				
GD	lr	0.003 $\rightarrow$ fl	0.01	0.03	0.1	
	recovery	0.113	0.131	0.437	div	
Adam	lr	0.0003 $\rightarrow$ fl	0.001	0.003	0.01	0.03
	recovery	0.560	0.576	0.570	0.573	0.581
scalar-Adam	lr	0.0003 $\rightarrow$ fl	0.001	0.003	0.01	
	recovery	0.165	0.201	0.256	0.357	
Muon (cosine)	lr	0.003	0.01	0.03	0.1	
	recovery	0.915	0.748	0.000	0.000	
Shampoo	lr	0.01	0.03	0.1	0.3	
	recovery	(0.881 n/i)	0.286	div	div	

Selection-rule stress test: with best-recovery selection under the uniform cosine schedule, the classification achieves 9/9; with fastest-interpolation selection it achieves 8/9, with *Muon* the sole failure at recovery 0.91. Thus the fastest-interpolation, bias-blind rule hurts an otherwise *equivariant* method, which is why Appendix D.2 reports results in terms of capability (can/cannot) with the curves, rather than under any single selection criterion. “Fastest-interpolating” is defined within the fixed horizon of 2×10^4 steps: the rate that crosses the 10^{-7} bar in the fewest steps, not the largest rate that eventually interpolates. For Muon that is the smallest grid rate (0.003, recovery 0.915 in Table 7); hence the two higher rates (0.03, 0.1), at which it recovers exactly, are not what this rule picks. The coordinate-wise cluster stays ≥ 0.42 under *both* rules.

D.4 INITIALIZATION SCALE

Table 8 sweeps the initialization scale for the five reference methods.

Table 8: Recovery vs. initialization scale (3 seeds). The split holds at 10^{-3} (the paper’s setting) and 3×10^{-3} ; at 10^{-2} the small-init bias fades for every method except Muon, which remains exact, marking its bias as schedule- rather than init-driven.

init	GD	Adam	scalar-Adam	Muon	Shampoo
10^{-3}	0.131	0.573	0.201	0.000	0.286
3×10^{-3}	0.203	0.575	0.260	0.000	0.352
10^{-2}	0.318	0.584	0.358	0.000	0.575

D.5 ATTENTION TWINS ACROSS SEEDS, INITIALIZATION DRAWS, AND NOISE LEVELS

Using the deterministic CPU setup described in §6, we trained six Adam gauge pairs—combining two different initialization seeds with three independent gauge draws. Initial drift for these runs varied from 3.4 to 5.4×10^{-3} , in good agreement with the same $2L/d=64$ configuration recalculated on GPU in float64, whose step-1 drift varied from 4.0 to 5.7×10^{-3} (Table 9). Final drift in this case ranged between 0.69 and 0.80 . The differences in validation accuracy were insignificant,

$|\Delta\text{acc}| \leq 0.0023$, suggesting that both models perform similarly well but converge to two different points on the solution manifold.

In the case of the Adam noise-twin tests performed in the same basis, noise amplitude varied from 10^{-7} to 10^{-5} , and the final drift in this case varied from 1.6×10^{-5} to 1.0×10^{-3} . The structural ratio, defined as the ratio of the *smallest* final gauge drift to the final drift of the noise twin carrying 100 times more perturbation, exceeds $600\times$, the actual run value being $662\times$. The rounded figure quoted here is the more cautious lower bound, and is the one we claim.

Among the equivariant rows, the SGD and scalar-Adam gauge twins, over two (seed, draw) combinations each, end at no more than 3.7×10^{-4} for SGD (its largest) and 4.3×10^{-6} for scalar-Adam.

The Muon chaos metric, the mean $|\log_{10}(\text{gauge}/\text{noise})|$ drift ratio past step 100, equals 0.05 decades for seed 42. Within that tolerance one may deem the gauge twin a perturbation at the floating-point level, and therefore indicative of chaos rather than of structural basis-dependence.

Scaling the twin diagnostic (GPU, float64). Table 9 reports analogous values for deeper and wider models, and on real text inputs. The $A=I$ determinism twin is exactly 0 for every row. Two observations are universal across all depths and widths: the initial Adam gauge deviation, which is about 10^{-3} – 10^{-2} , and the machine-precision “floor” shared by all three equivariant methods, which is 12–13 orders of magnitude below Adam’s value. The “ratio” column reports the “onset ratio”, defined as the smallest Adam gauge split at step 1 divided by the value of the $100\times$ -stronger noise twin at that same step. We report the onset, not the final, ratio, because in deep models the end-state statistics cease to be informative about the dynamics, even in fully deterministic, non-minibatch runs.

At $2L/d=64$ and $4L/d=128$, the equivariant twins remain negligible by the end—scalar-Adam at 6.6×10^{-15} and 6.0×10^{-12} , SGD at 2.5×10^{-9} and 1.6×10^{-10} —and the final-drift ratio is substantial (exceeding $500\times$ for $2L/d=64$; the CPU value mentioned above reflects the $662\times$ of the same comparison on that harness). At $6L/d=256$, however, the expectation is violated: scalar-Adam’s gauge twins produce a value of 1.8 and Muon’s 0.58, while SGD gives no signal at that scale, never leaving chance accuracy, so its exactly zero drift reports an untrained model rather than a preserved gauge. The noise twin interprets the rest as instability, with scalar-Adam’s noise twin at 1.7 and Muon’s at 0.83, corresponding roughly to where a 10^{-7} perturbation in the same basis would land at those scales. Similarly, the char-LM has scalar-Adam’s gauge and noise twins both at 0.44, while Adam is the only method with a discrepancy between its gauge and noise twin at step 1—which is why the step-1 onset, rather than the final state, carries the claim.

Table 9: Twin drift measured at scale and on real text, in GPU float64 precision. For each configuration, Adam runs 6 init $\times$ draw gauge pairs while each equivariant method runs 2; the $A=I$ determinism twin is evaluated for Adam at every configuration and is exactly 0 throughout. “eq. step 1” is the maximum (worst) gauge drift across the three equivariant methods (SGD, scalar-Adam, Muon) at step 1, all at machine precision; “ratio” is the onset ratio defined above. The character-level language model twins use the same deterministic minibatch stream and reach comparable validation losses (1.579 vs. 1.585).

config	Adam gauge, step 1	eq., step 1	Adam, final	ratio (step 1)
mod-47, 2L, $d=64$	$4.0\text{--}5.7 \times 10^{-3}$	5.4×10^{-16}	0.69–0.78	93 $\times$
mod-97, 4L, $d=128$	$8.2\text{--}11 \times 10^{-3}$	7.3×10^{-16}	0.66–0.75	158 $\times$
mod-97, 6L, $d=256$	$6.5\text{--}7.2 \times 10^{-3}$	1.1×10^{-15}	0.61–0.64	94 $\times$
text (char-LM), 6L, $d=256$	$2.2\text{--}2.4 \times 10^{-3}$	8.0×10^{-16}	0.36–0.37	39 $\times$

D.6 HYPERSPECTRAL LEARNING-RATE HYGIENE

The learning rates in §9 are chosen via a train-only criterion: the best rate is the one achieving the lowest training loss within the step budget, with ties broken in favor of the one using fewest steps. We used different seeds for selection than for two of the four evaluation seeds. Following this protocol, GD selects learning rate 10 at density 0.15, which matches the test-informed choice, but with no data leakage—this is the only rate on the grid that can reach training loss 3×10^{-5} in 3×10^4 steps (lr 3 reaches 3×10^{-4} , lr 1 only 3×10^{-3}).

A per-rate analysis using four different seeds confirms that this is not an edge-of-stability (Cohen et al., 2021) effect: GD with lr 3 tracks the same matched-loss trajectory as the lr=10 run—at every

comparable loss, the held-out RMSEs differ by at most 10^{-5} (e.g., both have 0.03487 at one point and 0.02207 later). At its deepest point achieved (train loss $\leq 3 \times 10^{-4}$, all four seeds), GD(lr=3) already outperformed Adam by 12%, and the gap widens to 44% at deeper levels with the higher learning rate. For the two rates tried on GD and the one density, the matched-loss path is therefore independent of the rate (within the rate class), varying only in how far along the path the optimization gets within the step budget.

Two limits of this grid. (i) Our reported improvement depends on the selection criterion, though not the ordering: train-only picks better rates for faster convergence, hence bigger rates for the same budget. In the GPU-replicated run (Table 5), whose grid extends to lr 30, this puts GD(lr=30) in its rate-invariant region while giving Adam the worst rate on its own grid. Adam’s smallest rate (10^{-3}) achieves held-out RMSE 0.0171 at effective rank 17, compared to GD(lr=30)’s 0.0148 at rank 13. So picking the best rate for each method would have improved GD by 13% over Adam, compared to the 43% found with the train-only selection (also consistent across the four paired seeds in the GPU experiment). We report the train-only selection because using held-out statistics to both select and evaluate a learning rate would reintroduce the very bias this protocol was designed to avoid (Appendix D.2). What does not change with the selection criterion, however, is the ordering and rank profile: at every rate on Adam’s grid (0.0171/17, 0.0198/20, 0.0260/27), the GD run selected achieves both a lower held-out error and lower effective rank than the corresponding Adam run.

(ii) Scalar-Adam is undersampled in these experiments: its rate grid $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ sits at $3 \times$ higher rates than Adam and shows no signs of saturating, achieving lower held-out error still at the smallest rate tried (0.0207 vs 0.0360 at the selected rate) and hence does not reproduce the phenomenon. This is entirely consistent with the finding in §8—equivariance defines the algorithm class, while spectral responses determine the dynamics within a class. Real-world data has spectral content that makes this second channel relevant, and scalar-Adam’s EMA of first moments (Remark 4.7) fails to satisfy the premise of Theorem 4.6. A similar consideration applies to Shampoo across problem sizes (Appendix D.9).

D.7 INTERPOLATION-BAR AUDITS

RMSProp necessitated the formulation of the rule in §3 based on performance criteria rather than method class, due to its behavior. Under a constant learning rate, it fails to interpolate, plateauing at a training loss of 2.9×10^{-4} . This limit is not due to insufficient training, as even with a threefold increase in training budget (up to 6×10^4 steps), the lowest achieved loss remains at 2.8×10^{-4} . The bottleneck stems from the ϵ -bounded denominator in its update rule, which persists regardless of extended training. Evaluating recovery performance at this plateau would mean comparing a non-interpolating, early-halted run with eight interpolating cases—precisely the kind of comparison the evaluation protocol aims to avoid.

However, when using cosine decay—better suited to RMSProp’s optimization dynamics—it successfully interpolates, achieving a training loss of 6.0×10^{-8} at a learning rate of 10^{-4} . In this setting, it reaches a recovery score of 0.5266 with an effective rank of 12.80. This result falls within the coordinate-wise cluster and lies just 0.0004 below the 0.5270 score of the constant-rate run—measured at its 2.9×10^{-4} plateau, since that run never met the interpolation bar, so 0.5270 is a recovery at the plateau, not at interpolation. Thus, the learning schedule has minimal impact on final recovery performance despite significant differences in training dynamics; it determines whether RMSProp is measurable, not what it selects. Once we apply decay, all nine methods satisfy the 10^{-7} interpolation threshold, giving rise to a 9/9 classification (Table 2). This conclusion holds true when every method follows the same cosine learning-rate schedule (Appendix D.1); thus, none of the clustering results presented here depend on a specific method’s scheduling scheme.

Conversely, our evaluation also turned up the opposite failure case. With a constant learning rate, the signum algorithm stagnates at 5×10^{-2} training loss, attaining a recovery score of 0.009. If recovery were measured without consideration of interpolation, this early-stopping behavior would be considered evidence in favor of sign descent being able to maintain bias. In contrast, with interpolation (under decay), signum has a recovery of 0.445, indicating that the bias was lost, as expected.

D.8 FLOWADAM-P IS NOT AN ARTIFACT OF EARLY STOPPING

The bias restoration achieved in §10 with FlowAdam- $p=0$ (using RMS convention, global norm clipping at $c=10$, and learning rate 10^{-3}) is subject to the same early-stopping ambiguity as any recovery measurement. In order to investigate this, we boosted the training budget to 1.2×10^5 steps and measured recovery at matched training loss points ranging from 10^{-4} to the stricter 10^{-7} interpolation threshold. Recovery stays relatively flat within this range (Table 10): each individual run varies by no more than 3×10^{-4} between 10^{-4} and 10^{-7} , and the values are stable to four decimal places beginning at 10^{-6} . Therefore, the recovery and effective rank obtained with a lax loss floor are nearly indistinguishable to three significant figures from those obtained at full interpolation. The bias is well formed before the final stop, demonstrating that it is not an artifact of the stopping rule.

Table 10: FlowAdam- $p=0$ recovery along a trajectory aligned by training loss (extended 1.2×10^5 -step budget, 3 seeds). The mean at the 10^{-7} threshold (0.1691) matches the original 3×10^4 -step result exactly and improves upon the dial-only baseline (0.2010) by +15.9%. Its effective rank (4.8–5.4) is the closest of any Adam variant in this study to gradient descent’s (4.51).

train loss $\leq$	10^{-4}	10^{-5}	3×10^{-6}	10^{-6}	10^{-7}	erank
seed 42	0.1404	0.1401	0.1401	0.1401	0.1401	4.77
seed 123	0.1558	0.1556	0.1555	0.1555	0.1555	4.84
seed 456	0.2120	0.2117	0.2117	0.2117	0.2117	5.41
mean	0.1694	0.1691	0.1691	0.1691	0.1691	5.01

D.9 THE H100 REPLICATION LADDER

The mechanism-identification experiments presented in the main text are intentionally compact and deterministic. To directly assess how results vary with scale, we evaluated the full suite of methods on the sensing task across problem sizes $n \in \{64, 96, 128, 192, 256\}$ (with planted rank 3, $m = 2$ dof, using `float64` precision on H100 GPUs), using 10 random seeds per configuration. For each method, three seeds are used to choose the learning rate based on the best average recovery performance among the interpolating rates; see Appendix D.3, where this selection criterion is rigorously tested. The other seven seeds are used only with the selected rate and not in selection; averages presented here are over all 10 seeds. Importantly, the seven holdout seeds alone recapitulate the same two-cluster ranking order at each problem size; e.g., at $n=256$, $\max(\text{GD}, \text{sc-Adam}, \text{Muon})=0.56 < 0.71 = \min(\text{clean coordinate-wise})$ vs. $0.55 < 0.71$ using all the seeds in Table 11 (in which “sc-Adam” means scalar-Adam and “clean” means excluding the flagged $\text{signum}^\ddagger$). The training budget is fixed at 4×10^4 steps.

Learning rate grids for each problem size are scaled so that each method operates within the interpolating regime across all sizes: GD scales by $40/n$; Muon, Shampoo, and ScaledGD by $\sqrt{40/n}$, with ScaledGD included as a schedule control but not reported; Adam-family methods scale by $\sqrt{n/40}$. At $n=40$, these grids match exactly those used in the main paper, serving as a validation check. A cosine-annealing control was also run at each size. All methods yield nearly identical recovery values under both the constant-step and cosine schedules—up to three decimal places—with one exception: RMSProp, whose selected rate becomes unstable at $n=64$ and $n=96$ (recovery shifts from 0.596 to 0.610 and 0.615 to 0.767 under cosine), though it remains within the coordinate-wise cluster regardless. Thus, the relative ordering in the ladder is unaffected by the choice of schedule. The ladder stops at $n=256$: $n=384$ is omitted because boundary effects make the grid uninformative there, with Muon no longer interpolating.

Discussion around the ladder. The ranking in Table 11 is what underlies our conclusion, and one number in it is easy to misread: the Adam-family ratio with respect to GD decreases with growing n , but this is not because Adam is relatively better, rather because GD’s absolute value climbs toward its flow-limit value under the same step budget. Both GD and scalar-Adam would approach the continuous-time flow limit given smaller steps and more of them, which is why the ordering rather than the absolute error is what the table is asked to carry.

The $n=40$ canary run correctly recovers the results of the paper when the protocol is followed, giving 0.424 for Lion (versus 0.425 in the paper) and 0.543 for Adafactor (matching the paper’s

Table 11: Optimization algorithms on the Zoo recovery task, across problem sizes (10-seed averages; float64, H100 GPU, 4×10^4 steps). The ranking is consistent across scales: gradient descent (GD), scalar-Adam ($p=0$), and Muon significantly outperform coordinate-wise Adam, RMSProp, Lion, and Adafactor, with Adam-type methods at $1.7\text{--}4.3\times$ GD’s error at every size and Muon almost exact (error $<10^{-5}$ for all seeds) up to $n=256$. The errors of GD and scalar-Adam are limited by the step-size choice (Remark A.5), so the relative ordering, not the absolute value, is the conclusion.

n	equivariant				coordinate-wise				
	GD	sc-Adam	Muon	Shampoo [†]	Adam	RMSProp	Lion	Adafactor	signum [‡]
64	0.143	0.195	0.000	0.646	0.574	0.596	0.483	0.612	0.000
96	0.202	0.279	0.000	0.595	0.609	0.615 [§]	0.548	0.687	0.000
128	0.248	0.345	0.000	0.659	0.643	0.617	0.602	0.745	0.000
192	0.346	0.468	0.000	0.838	0.696	0.669	0.760	0.817	0.000
256	0.410	0.548	0.000	0.918	0.741	0.708	0.895	0.867	0.000

[†]Shampoo’s damping $\lambda=1$ is fixed independently of problem size, which is why it does poorly at large n —a hyperparameter choice, not an intrinsic limitation. [‡]signum reaches comparable performance on a sufficiently long training schedule, through a different mechanism (Appendix D.10). [§]RMSProp has one outlier at $n=96$, where a poorly chosen learning rate makes it diverge on some seeds.

0.543), while Adam stays flat across the extended grid, registering 0.550 at its finest interpolating rate. Meanwhile, both GD and scalar-Adam achieve better recovery with smaller step sizes, reaching 0.108 and 0.143 respectively on their finest grids (GD’s 0.108 cell stops at a mean training loss of 1.1×10^{-7} , just above the bar, so we read it as a flow-limit value rather than an interpolating one)—agreeing with the flow-limit behavior described in Remark A.5, which is why the larger- n values on the ladder should be considered conservative. In particular, two cases were kept in the table for transparency rather than optimized out: Shampoo with untuned damping, and signum’s long-horizon recovery. Neither affects the stated ranking, and both are flagged for further investigation (§11).

D.10 RECOVERY WITHOUT EQUIVARIANCE: ANNEALED SIGN DESCENT AT LONG HORIZONS

The zoo’s two-cluster separation (§5) is defined under a fixed computational budget. However, one coordinate-wise method manages to escape the collapsed cluster when training is extended: signum (signSGD with momentum) using cosine decay over a prolonged annealing schedule. At the paper’s standard budget of 2×10^4 steps, its recovery score is 0.445—classified as collapsed. When extended to the ladder’s 4×10^4 -step horizon, recovery drops to near-zero (below 10^{-5}) for all $n \geq 64$ (across all 10 random seeds). The transition begins at $n=40$, a borderline scenario depending on numerical precision and the random seed. On H100 in float64, two of the three seeds recover, for a mean error of 0.013; the original CPU-based implementation (optimizer_zoo_bias.py, MAX_STEPS= 4×10^4 , lr 3×10^{-3}), meanwhile, gives a mean of 0.054, with the run using seed 42 recovering to 1.5×10^{-5} ; and at lr 10^{-3} , recovery stays poor at about 0.71 throughout.

The values shift between individual runs and platforms, but the configuration as a whole—large step sizes and an extended annealing schedule—reproduces the effect. It is not random, nor is it due to numerical instability: we have observed the phenomenon but do not yet have a mechanistic account of it, so we report it as an empirical fact.

The example does not disprove the theorem, but instead demonstrates where it does not apply. Theorem 4.6 concerns flows with memoryless updates and a scalar preconditioner shared across the factors, and does not consider alternatives. It is entirely possible that a different set of update rules, not respecting the equivariance of the loss under the gauge, would have different dynamics and still be able to exploit the low-rank structure of the target signal. We considered one possibility and ruled it out.

A possible explanation could be that signum’s momentum has an intrinsic rank-one structure: if the momentum were an outer product, then since $\text{sign}(\sigma uv^\top) = \text{sign}(u)\text{sign}(v)^\top$ for $\sigma > 0$, the signed update would itself be rank-one and aligned with the planted signal. This does not appear to be the case. At 4×10^4 steps, the successful runs at lr 3×10^{-3} have a ratio of the top two singular values of the momentum of $\sigma_1/\sigma_2 \approx 1.1\text{--}1.3$ at convergence, and the signed updates themselves have even lower values (1.05–1.17). The fraction of squared Frobenius norm contained in the leading singular component—the top-mode energy—lies between 10% and 24% for these runs (experiments/signum_c9_audit.py), against the 100% a genuinely rank-one matrix would carry.

The leading mode therefore holds only a small share of the update’s energy, so the update as a whole cannot have been rank-one. The mechanism by which signum achieves lower-rank recovery after long annealing schedules remains an open question.

A more plausible theory concerns the ℓ_∞ or steepest-descent nature of sign-based updates (Bernstein et al., 2018; Xie & Li, 2024), which implicitly biases towards max-margin solutions—Baek et al. (2026) prove that signum converges to the ℓ_∞ -max-margin classifier at any batch size. These, in turn, have low rank only if the ground truth is also low-rank, which would explain the collapse of the corner in the presence of any nontrivial spectral tail (as illustrated below); we posit this as an open problem.

Finally, this example differs from the common-scalar case in that it does not appear to stem from a continuous limit. First, it emerges from large learning rates and long annealing schedules rather than from a smooth flow limit. Specifically, at a learning rate of 3×10^{-3} , the average recovery over three random seeds improves from $0.445 \rightarrow 0.054 \rightarrow 0.000$ as the cosine-annealing schedule is extended from $2 \times 10^4 \rightarrow 4 \times 10^4 \rightarrow 6 \times 10^4$ steps. By contrast, at a learning rate of 10^{-3} , it remains poor (0.65–0.72) throughout the schedule. A linear decay schedule has a similar threshold behavior ($0.479 \rightarrow 0.103 \rightarrow 0.000$ at analogous durations), which suggests that the issue is not specific to cosine annealing.

Second, the corner structure breaks down when a spectral tail is added to the planted matrix. In the tail sweep described in §8, signum at 3×10^{-3} and 4×10^4 steps shows recovery values of 0.025, 0.138, 0.280, 0.422, 0.646, and 0.796 at $\tau = 0, 0.05, 0.10, 0.20, 0.35$, and 0.50 respectively (averaged over three seeds; the $\tau=0$ value again reflects the seed-sensitive borderline case, and sign descent is chaotic enough here that these digits shift by up to 0.03 across platforms while the monotone breakdown does not). Meanwhile, common-scalar methods follow the smoother degradation patterns seen in Table 4. Thus, we view signum as an exceptional case that helps define boundaries: non-equivariant dynamics can achieve low-rank solutions only in a narrow regime—exactly low-rank settings, large steps, and long anneals—while common-scalar equivariance supports a more robust, generalizable, and analyzable gradient-flow mechanism.

E EXTENDED RELATED WORK

Transformation invariance in optimizer design. The engineering literature has grappled with similar design choices at a lower level; for instance, Ling et al. (2022) observe that for vector-valued geometry parameters Adam’s per-coordinate moment estimation incorrectly violates rotation equivariance, and restore it by using a single shared scalar per vector. This is an early optimizer-level analog of the continuous transformations that our dial (§7) performs. Recently, LoRA-RITE (Yen et al., 2025) proposes a transformation-invariant matrix preconditioner on LoRA (low-rank adaptation, Hu et al. 2022) parameters, which improves practical efficiency by explicitly accounting for non-invariance as a source of optimization error. We import this line of reasoning to the implicit-bias setting, showing that failure to respect invariance not only slows down training but biases the solution, in the factored parameterization that a LoRA adapter instantiates (§5) and in standard attention layers (§6). Riemannian and preconditioned optimizers such as ScaledGD (Tong et al., 2021) incorporate similar domain knowledge for better conditioning; ScaledGD is also our within-class counterexample (Remark 4.7).

Independently, Zhang (2026) discovers a similar local phenomenon in a different application domain: in group-equivariant networks, Muon’s polar step respects orthogonal transformations in the Schur multiplicity basis while AdamW’s coordinate-wise preconditioning does not; this leads to a new optimizer, PolarAdamW, which deliberately breaks equivariance to obtain per-coordinate adaptation. These works share the starting point with us in that they identify a local covariance structure that should be respected by the optimizer, but we are, to our knowledge, the first to analyze its consequences for solutions recovered by factored models. Their observation that coordinate-wise preconditioning is most useful when the multiplicity gauge is trivial corresponds to our boundary analysis (§10.1).

Recent works explore similar themes from different angles: Lau & Su (2026) propose that update directions should respect the symmetry group of each weight block, unify spectral descent, Muon, Scion, and their polar variants, and derive equivariant optimizers for permutation and shift symmetry. Shirodkar (2026) generalizes standard optimizers to be exactly G -equivariant across architectural

gauges, e.g., rotations within each attention head; the outcomes it reports—an Adam variant that resists the over-training collapse AdamW falls into, and a Muon variant that groks at a depth plain Muon never reaches—are selection effects of the kind our mechanism predicts, arrived at from the design side. [Stupariu & Manolache \(2026\)](#) directly compare Muon and Adam in equivariant settings. We view these works as providing important context and design inspiration; in particular, by analyzing the interaction between optimization and symmetry we highlight cases where breaking equivariance (§10.1) is desirable.

Symmetry-driven accounts of implicit bias. Accounts based on symmetry offer a framework for understanding implicit bias. [Aladrah et al. \(2026\)](#) propose a complementary stochastic mechanism in which continuous predictor-preserving symmetries interact with noise from stochastic gradient descent (SGD), leading to a geometric correction—based on quotient volume—to the effective loss. This adjustment accounts for balancing and spectral effects and enables reverse engineering of the bias. Their illustrative case involves positive rescaling symmetry (UD, VD^{-1}). In contrast, our approach is deterministic and tied to specific optimizers rather than driven by noise or applicable universally across them: full-batch update rules either commute with or explicitly depend on the orthogonal gauge, and this property alone determines which interpolant is selected under a fixed computational budget (as verified in our bitwise-deterministic twin experiments). These two mechanisms are treated as separate in our analysis, and how they might interact under mini-batch training remains an open question noted in §11.

Muon and structured optimizers. Regarding structured optimizers, Muon ([Jordan et al., 2024](#); [Liu et al., 2025](#)) applies matrix-sign orthogonalization to momentum, while Shampoo ([Gupta et al., 2018](#)) uses Kronecker-factored preconditioning—both methods seeing practical use. Recent theoretical work attributes a spectral max-margin bias to Muon ([Fan et al., 2025](#); [Gronich & Vardi, 2026](#)), and [Kang et al. \(2026\)](#) establish equal-rate singular-value growth for Muon when applied to LoRA-style factorizations. Our phase diagram captures these dynamics, predicting when Muon’s scheduling is beneficial (for exactly low-rank targets) versus detrimental (for targets with heavy spectral tails), aligning with the dual empirical behavior observed by [Dragutinović et al. \(2026\)](#).

Independently, [Beneventano et al. \(2026\)](#) identify the polar step as the entropy-maximizing (spectrum-flattening) choice in one step and derive exact singular-value dynamics in underdetermined regression, providing theoretical grounding for the tail-sensitive mechanism measured in our phase diagram. [Shen et al. \(2026\)](#) arrive at a consistent view regarding performance trade-offs in mixed-spectrum sensing tasks. [Dong & Sawin \(2026\)](#) directly implement a within-class family by raising gradient singular values to a fractional power, effectively interpolating between Muon and gradient descent along the spectral-transfer axis identified in our structure theorem (§4) and visualized in our phase diagram. Our two axes place all of these on one map.

F RELATION TO INVARIANT-OPTIMIZER DESIGN

LoRA-RITE ([Yen et al., 2025](#)) shows that invariant preconditioners are practical, and our results supply a reason to want them beyond optimization efficiency: basis-dependence changes what is learned. The dial’s $p=0$ endpoint and global-norm clipping are two minimal repairs that are already implemented (intermediate p tempers the broken symmetry without restoring it), and full matrix preconditioners (Shampoo, Muon, RITE-style) are the aggressive end of the same axis.